



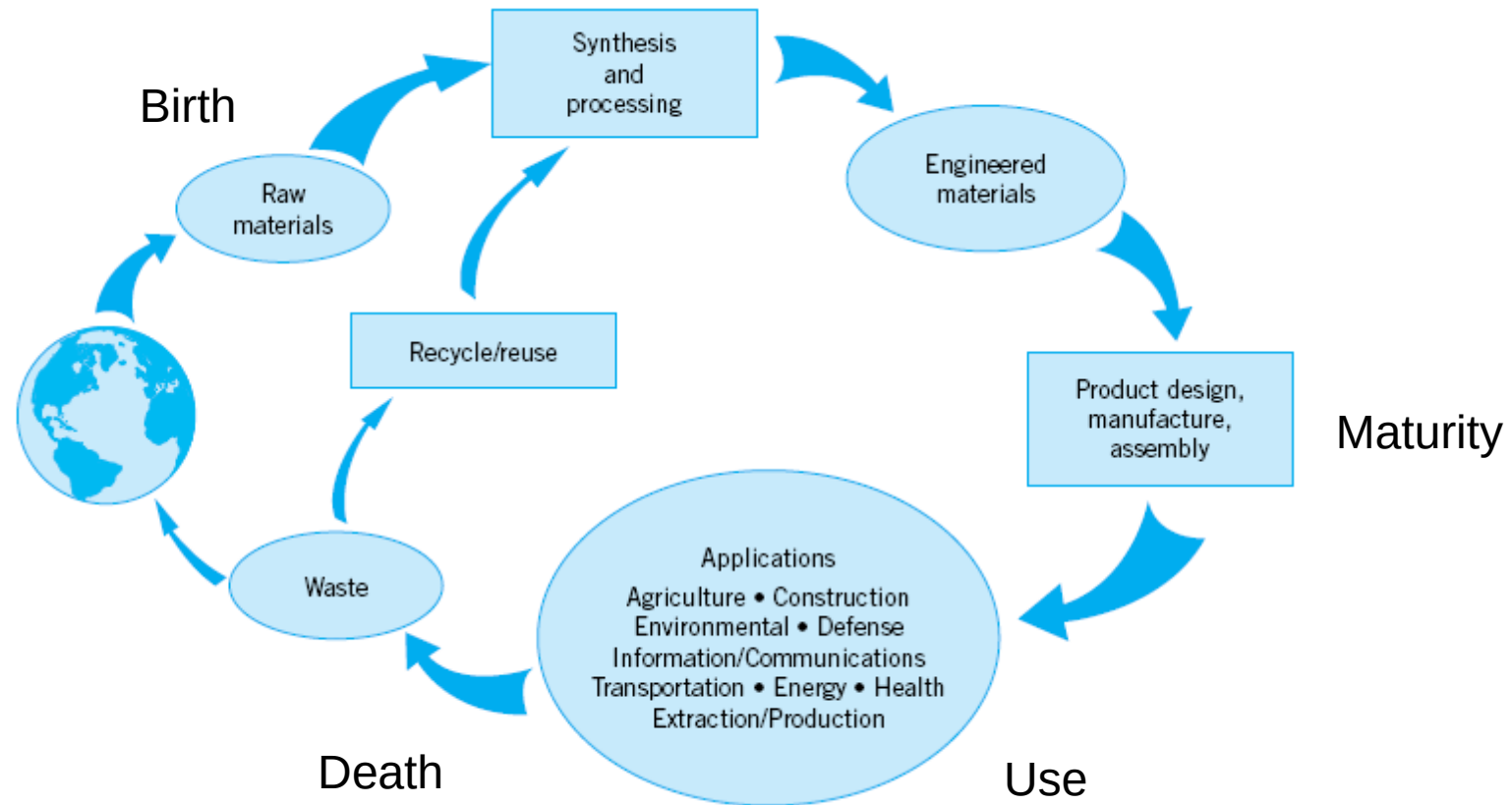
Life Cycle Assessment

ENGEN370 – Mark Lay



Product life cycle

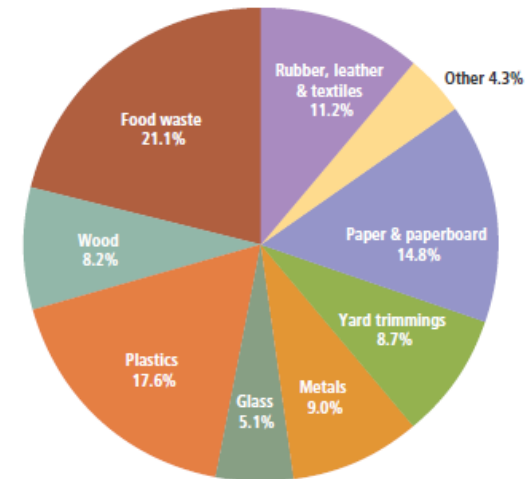
- 15 billion tonnes of raw material extracted from earth each year



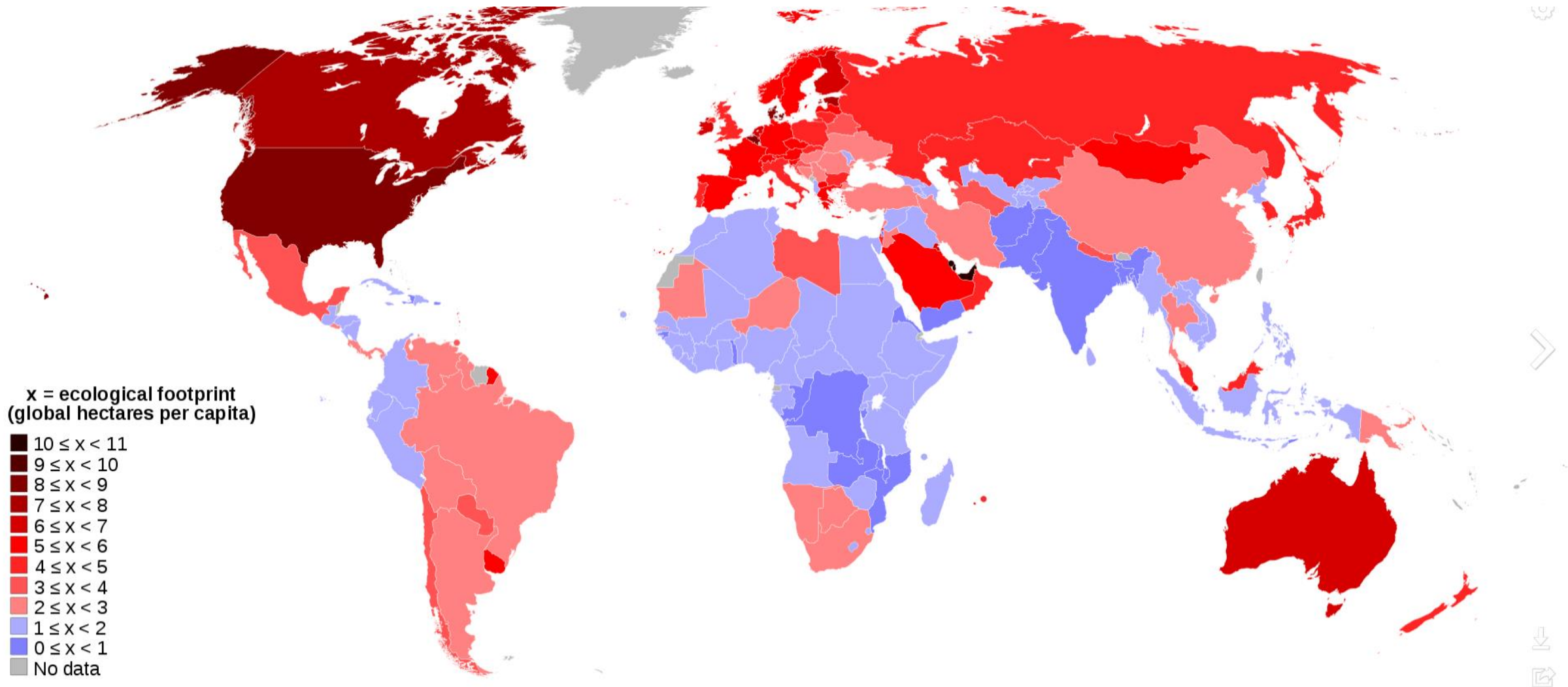
Municipal solid waste

- US – 250 million tonnes per year
 - 65 million tonnes recycled
 - 21 million tonnes composted
 - 29 million tonnes combusted
- NZ – 2.5 million tonnes per year
 - 75% could be potentially recycled, recovered, reused
 - 2008 Waste Minimization Act

Figure 7. Total MSW Discards (by material), 2012
164 Million Tons (after recycling and composting)

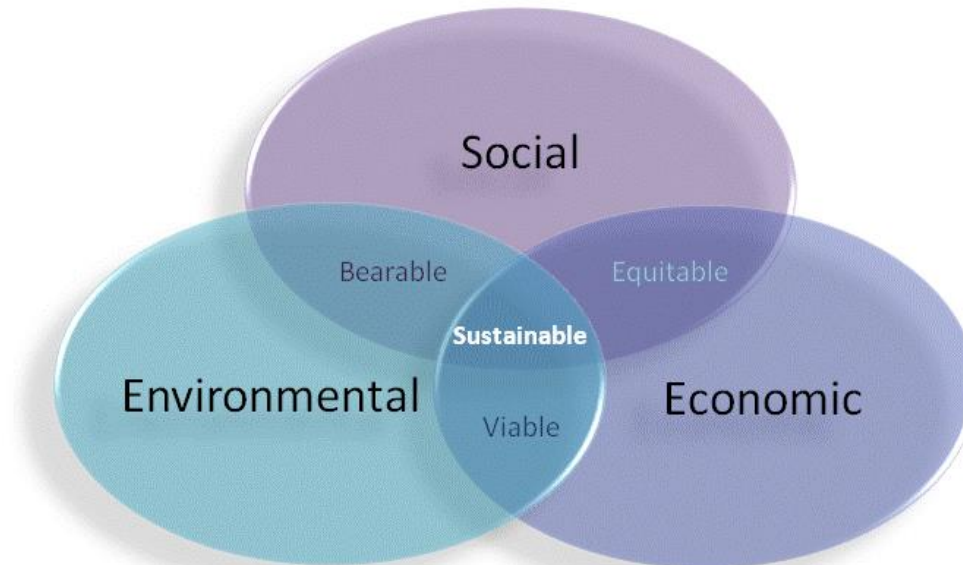


Ecological Footprint – 1.7x Earth



Sustainable Development

- Preserve genetic diversity
- Maintain agricultural, marine and forest systems
- Sustainable use of renewable resources
- Avoid degrading physical resources





- Set in 2015
- 193 countries

ISO 14040 - LCA

- Technique for assessing environmental and potential impacts associated with a product or service
 - Compiling an inventory of inputs and outputs of a product system
 - Evaluating impacts of those inputs and outputs
 - E.g. Energy demand, GWP, CO₂, CO, VOC, NO_x, SO_x, PM, photochemical oxidant potential, acidification, eutrophication, toxicity
 - Interpreting the results in relation to the objectives of the study
- Cradle to grave; cradle to gate; gate to gate; gate to grave
 - Cradle – birth of a material
 - Grave – death of a product and disposal
 - Gate – a stage in the production of a product

LCA

- Why is it useful?
 - Identify opportunities to reduce environmental impact – e.g. the tea bag
 - Decision making in industry, government or non-governmental organisations
 - Selection of relevant indicators of environmental performance
 - Marketing
- One of a set of environmental management techniques
 - Risk assessment
 - Environmental performance evaluation
 - Environmental auditing
 - Environmental impact assessment

Humble Tea Bag



LCA

- Can routinely use to improve products to make them more sustainable
 - Phillips
 - Unilever
 - Proctor and Gamble



LCA

- Equivalents
 - Mass and energy and whole life cost accounting for manufacturing processes
 - Cost accounting for different management/governmental strategies
 - E.g. whole life cost of weapons systems including use, maintenance and disposal
 - Cost benefit analysis for maximizing welfare – 1971 US executive Order 20503 – Quality of Life

First known LCA

- Coca Cola – 1969
 - Environmental impact of packaging
 - Resource use and disposal
 - Recyclable materials – steel, aluminium, glass and plastic (PET)
 - 2000 ultraglass contour bottle
 - 40% stronger, 20% lighter, 10% less expensive
 - Eliminated 52,000 tons of glass
 - CO₂ reduction of 26,000 tons per year
 - Reducing packaging weight



First links to public policy

- Netherlands – 1980s
 - Department of Environmental Management led by Peter Winsemius
 - Looked at water, air, soil
 - Acidification, eutrophication, diffusion of toxic substances, waste disposal, noise, odour, air pollution
 - Dutch companies Unilever and Philips routinely use it

Standardisation of LCA

- 1990's – Society of Environmental Toxicology and Chemistry (SETAC)
 - Brought LCA practitioners together
 - Harmonized methods, frameworks, practices
- 1994 - ISO – formalized standards, been ongoing since

LCA

- Limitations
 - Does not address economic or social impacts
 - Choices and assumptions made in carrying out the LCA are subjective
 - E.g. economic vs mass allocation, allocation of impacts to wastes?
 - Depth and detail will vary depending on goal and scope of the LCA
 - Models used are limited by their assumptions
 - Steady state, but what about Bhopal, Chernobyl, Fukushima?
 - LCA results may not be applicable for other settings
 - NZ (80% renewable) vs Aus (14.6% renewable) electricity
 - Accuracy of results dependent on quality, accessibility of data
 - Data not available, so have to make assumptions or find other sources to estimate
 - Temporal or spatial impacts not accounted for
 - Point source emissions, midnight valve, sudden spikes in emissions e.g. fire

ISO 14040

- ISO – International Organisation for Standardisation
- Membership of 161 national standards bodies
- Collaborates with 700 international, regional and national organisations
- Financed by subscriptions
- Based in Geneva, Switzerland
- Published 22136 International Standards
- Standards New Zealand is the NZ member for ISO

ISO 14000 family

- ISO 14040 – Life cycle assessment – Principles and framework
- ISO 14044 - Life cycle assessment – Requirements and guidelines
- ISO 14045 - Eco-efficiency assessment of product systems - Principles, requirements and guidelines
- ISO 14046 - Water footprint - Principles, requirements and guidelines
- ISO 14047 - Life cycle assessment - Illustrative examples on how to apply ISO 14044 to impact assessment situations

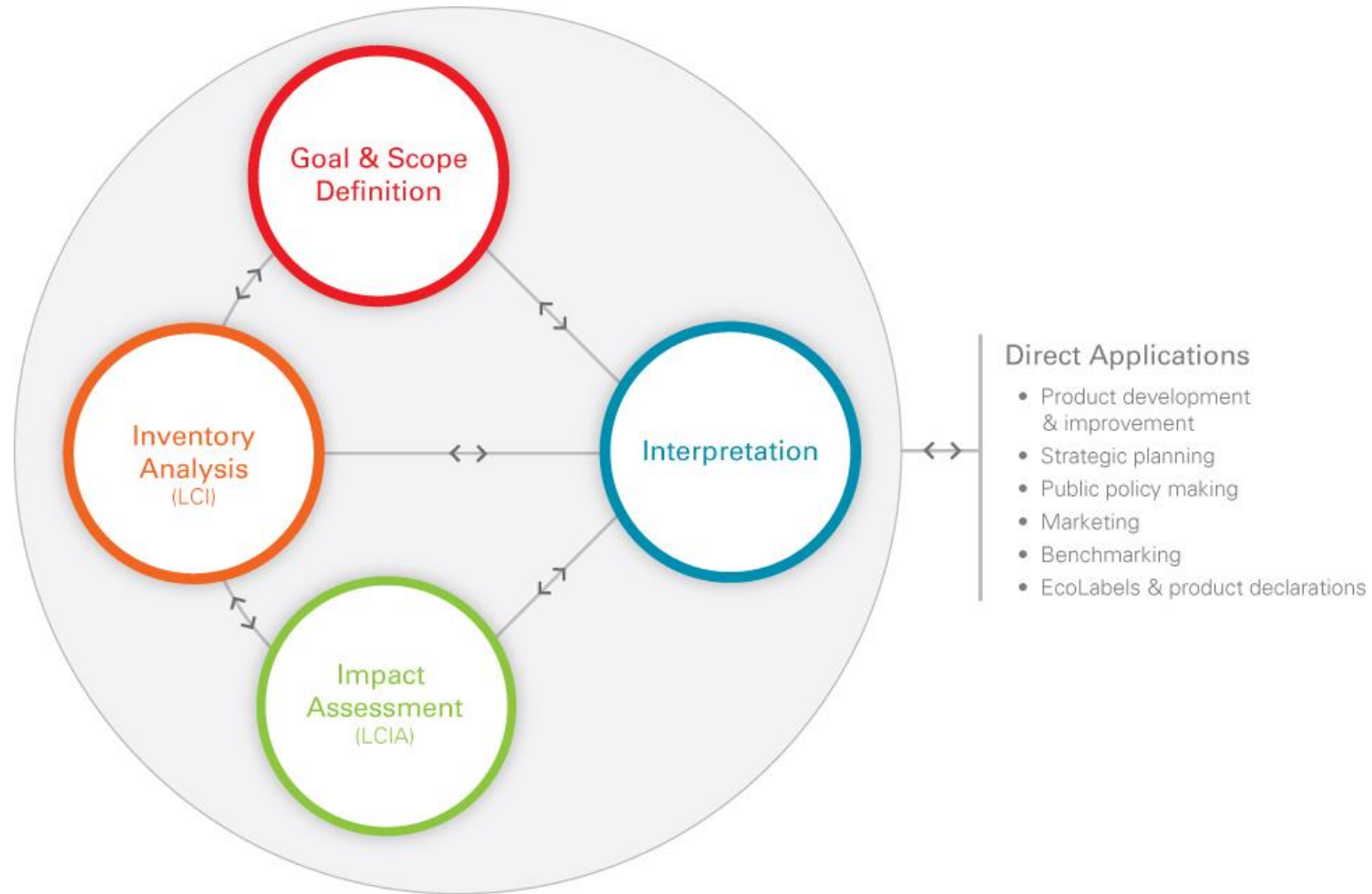
ISO 14000 family

- ISO 14048 - Life cycle assessment - Data documentation format
- ISO 14049 - Life cycle assessment - Illustrative examples on how to apply ISO 14044 to goal and scope definition and inventory analysis
- ISO 14071 - Life cycle assessment - Critical review processes and reviewer competencies: Additional requirements and guidelines to ISO 14044:2006
- ISO 14072 - Life cycle assessment - Requirements and guidelines for organizational life cycle assessment
- ISO 14073 - Water footprint - Illustrative examples on how to apply ISO 14046

LCA

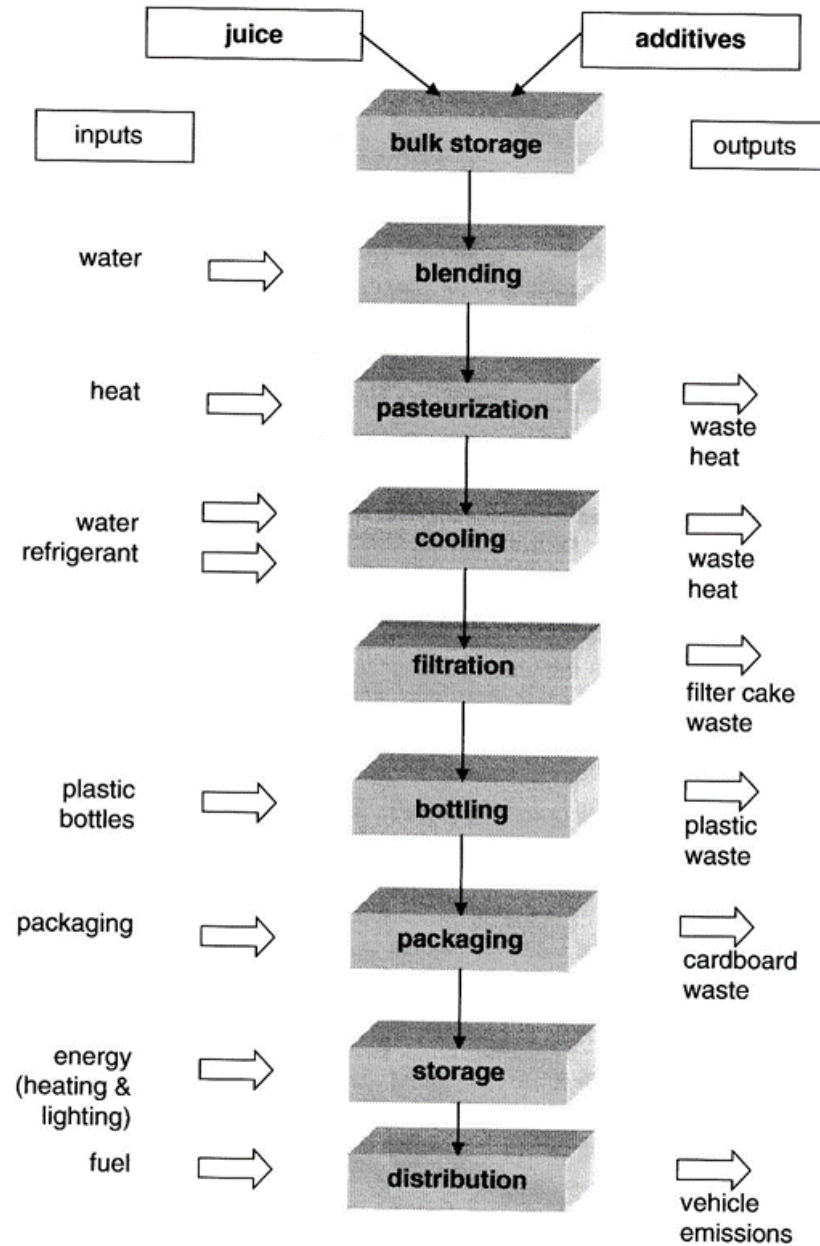
- Choices, assumptions, goal and scope need to be transparent and clearly documented
- Limitations and caveats should be stated
- Data sources should be clearly documented and referenced
- Model used should be clearly documented

LCA Phases - Iterative



LCA goal and scope

- Goal
 - Intended application
 - Reasons for the study
 - Intended audience
- Scope
 - Functions of the system being studied
 - Functional unit
 - Product system being studied
 - Product system boundaries
 - Allocation procedures



Ref: ISO 14001:
Environmental Systems
Handbook

LCA goal and scope

- Scope
 - Types of impact
 - Method of impact assessment
 - Interpretation
 - Data requirements
 - Assumptions
 - Limitations
 - Data quality requirements
 - Critical review
 - How it is reported and format of the report

Function and Functional Unit

- E.g.
 - Product – ton of product
 - Highways
 - Road – km of road
 - Road use – km travelled
 - Subdivision, housing - land – hectare or m^2 of land
 - Water treatment - water – m^3 of water
 - Excavation equipment
 - Tons or m^3 of earth moved
 - TV
 - Product – per screen
 - Use – hours used

Inventory analysis/impact assessment

- Break down into processes involved in manufacture and/or use, disposal
 - Gets large!
 - Decide which processes you want to examine – system boundaries
- Quantify material and energy inputs and outputs for each process
- Allocating an impact to each of those material and energy inputs and outputs
 - For material coming from outside the system boundary, impacts may be unquantified – system expansion
 - Multiple products from the same process
 - Weight impact – mass basis or economic basis

Impact data for materials/processes

- soca
- EuGeos' 15804-IA
- NEEDS
- ecoinvent
- PSILCA
- ESU World Food
- GaBi
- ELCD
- LC-Inventories
- Social Hotspots
- ProBas
- bioenergiedat
- Agribalyse
- USDA
- Ökobaudat
- Agri-footprint
- Comprehensive Environmental Data Archive (CEDA)

Impact data for materials/processes

- Journal papers, technical reports
- Data may be specific for region, country, process
- Calculations

Impact assessment

- Selection of impact categories
- Tally up each impact type to give an overall impact
 - E.g CO₂ per km of road

Interpretation

- Discussion/implications
 - What do the results mean in terms of the goal and scope of the study?
 - Any issues
 - Data completeness, sensitivities, uncertainties

Types of LCA

- Attributional – Looking at the impact of one system only (what we are doing)
- Consequential – Comparing effects of decisions made
 - Choosing between two or three methods of asphalt manufacture
 - Choosing between using recycled concrete of new concrete
- Process LCA – (what we are doing)
- Economic Input Output LCA
 - Aggregates the processes into commodities
 - Environmental impacts of each economic activity
- Hybrid Process LCA, EIO LCA
- Dynamic LCA – includes time dependent, temporal impacts



Life Cycle Assessment 2

ENGEN370 – Mark Lay



LCA Quarry

Quarry production: 3200 tons gravel per day

Functional unit: ton of gravel

Equipment used	Quantity	Hours used per day	Fuel consumption per unit (L/hr)	Electricity consumption (kWhr/ton)
Hydraulic drilling rig on excavator	1	2	10	
20 ton excavator with 3 ton bucket	2	10	20	
Cat 936E loader with 9 ton bucket	2	10	13.2	
40 ton hauler	8	10	27	
Crusher		10		2.2
Classification		10		1
Belts		10		0.1

Fuel type	Density (kg/L)	CV (MJ/kg)	Formula	MW (g/mol)	Mols CO2 produced /Mol Fuel	Combustion (kg CO2 /kg fuel)	Production (kg CO2 /kg fuel)	Total (kg CO2 /kg fuel)
Diesel	0.832	44.8	C12H24	168.96	12	3.13	0.096	3.22
Biodiesel	0.875	37.27	C19H34O2	295.36	19	2.83	-1.53	1.30
Methane		52	CH4	16	1	2.75	0.01	2.76
Coal		33	C8.8H0.5O0.3	116.92	8.8	3.31	0.01	3.32

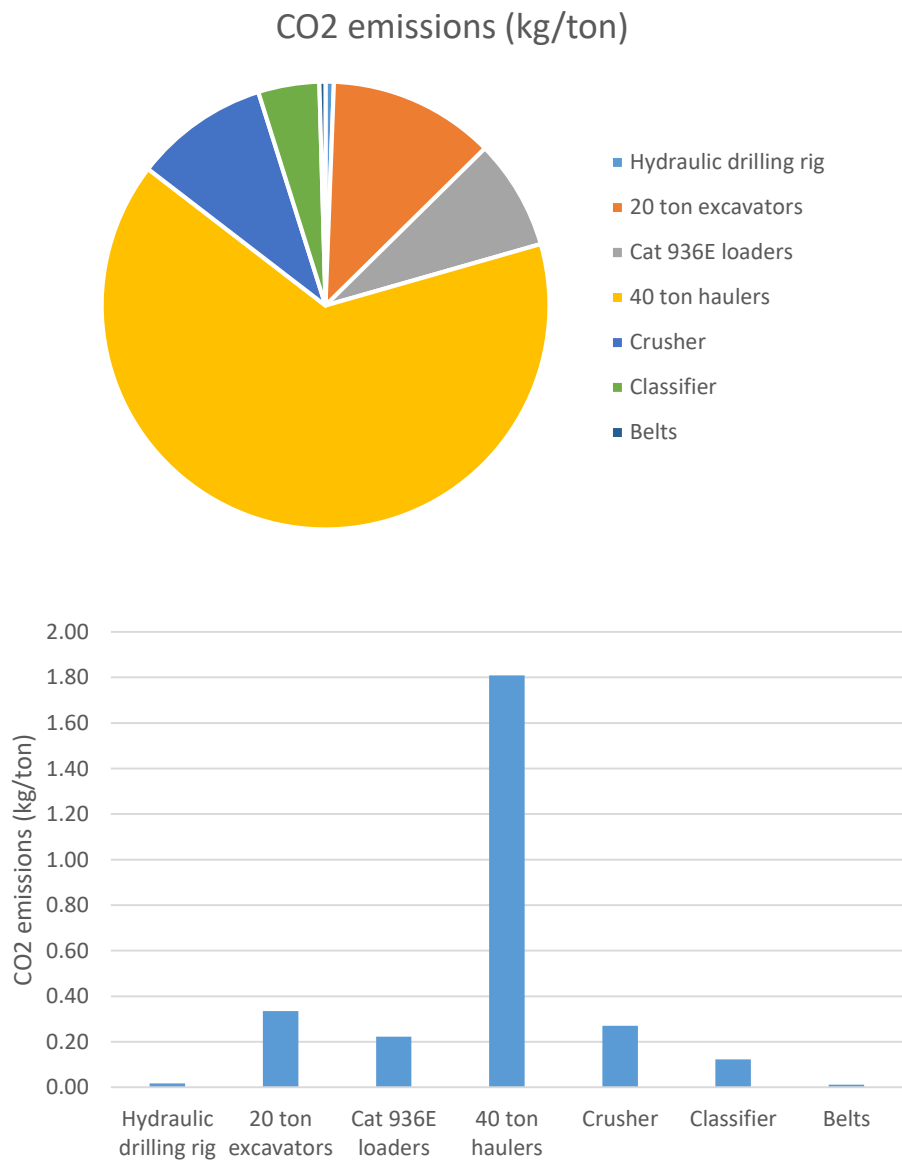
LCA Quarry

Huntly Power Station	Power (MW)	Efficiency	Capacity (%)	Actual (MW)	Thermal (MW)	Fuel type	Fuel consumption (kg/s)	CO2 emissions (kg/s)	CO2 emissions (kg/MJ)
Overall	900		65	585					
CC	400	0.56	100	400	714	Methane	13.7	37.9	0.095
Rankine	500	0.3	37	185	617	Coal	18.7	62.1	0.336
Total				585				100.0	0.171

Fraction renewable energy NZ = 0.8

LCA Quarry Mass Allocation

	NRE (MJ/day)	NRE (MJ/ton)	CO2 emissions (kg/day)	CO2 emissions (kg/ton)
Hydraulic drilling rig			53.6	0.02
20 ton excavators			1071.9	0.33
Cat 936E loaders			710.1	0.22
40 ton haulers			5788.5	1.81
Crusher	5068.8	1.584	866.3	0.27
Classifier	2304	0.72	393.8	0.12
Belts	230.4	0.072	39.4	0.01
Total	7603.2	2.376	8923.7	2.79



LCA Economic Allocation

NRE (MJ/\$)	CO2 (kg/\$)
0.101	0.119

Product	Price (\$/tonne)	Fraction	Mass (tonnes)	Revenue (\$)	NRE (MJ)	CO2 (kg)	NRE (MJ/tonne)	CO2 (kg/tonne)
Topcourse Domestic 20mm	15.8	0.015	48	758.4	76.65	89.96	1.60	1.87
Topcourse Standard 20mm	20.64	0.025	80	1651.2	166.87	195.85	2.09	2.45
Basecourse Domestic 40mm	13.37	0.025	80	1069.6	108.10	126.87	1.35	1.59
Basecourse Standard 40mm	17.69	0.025	80	1415.2	143.02	167.86	1.79	2.10
Road Metal 65mm	11.54	0.15	480	5539.2	559.80	657.02	1.17	1.37
Road Metal 65mm M3 (NZTA Specification)	19.85	0.3	960	19056	1925.83	2260.29	2.01	2.35
Bedding 7mm	19.24	0.05	160	3078.4	311.11	365.14	1.94	2.28
Bedding 10mm	22.99	0.05	160	3678.4	371.75	436.31	2.32	2.73
Bedding 14-5mm	27.36	0.05	160	4377.6	442.41	519.24	2.77	3.25
Bedding 20mm	27.36	0.05	160	4377.6	442.41	519.24	2.77	3.25
Filter B 100-250mm	32.07	0.08	256	8209.92	829.71	973.80	3.24	3.80
Cribwall Backfill 40-20mm	27.24	0.03	96	2615.04	264.28	310.18	2.75	3.23
Cribwall Backfill 80-40mm	21.92	0.03	96	2104.32	212.67	249.60	2.22	2.60
Sealing Chip Grade 2	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 3	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 4	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 5	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 6	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Concrete Aggregate 14-5mm	39.7	0.01	32	1270.4	128.39	150.69	4.01	4.71
Concrete Aggregate 20-10mm	38.7	0.01	32	1238.4	125.15	146.89	3.91	4.59
Total		1	3200	75233.28	7603.20	8923.66	2.38	2.79

LCA Sensitivity

Fuel consumption	L/hr	Lower (L/hr)	Upper (L/hr)	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
20 ton excavator	20	18	22	2.79	2.75	2.82	-1.43	1.08
Cat 936E loader with 9 ton bucket	13.2	11.9	14.6	2.79	2.77	2.81	-0.72	0.72
Hydraulic drilling rig	10	9	11	2.79	2.79	2.79	0.00	0.00
40 ton hauler	27	24.3	29.7	2.79	2.61	2.97	-6.45	6.45

Electrical	kWhr/tonne	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Crusher	2.2	1.98	2.42	2.79	2.76	2.82	-1.08	1.08
Classifier	1	0.9	1.1	2.79	2.78	2.8	-0.36	0.36
Belts	0.1	0.09	0.11	2.79	2.79	2.79	0.00	0.00
	Fraction	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Renewable electricity	0.8	0.72	0.88	2.79	2.95	2.63	5.73	-5.73

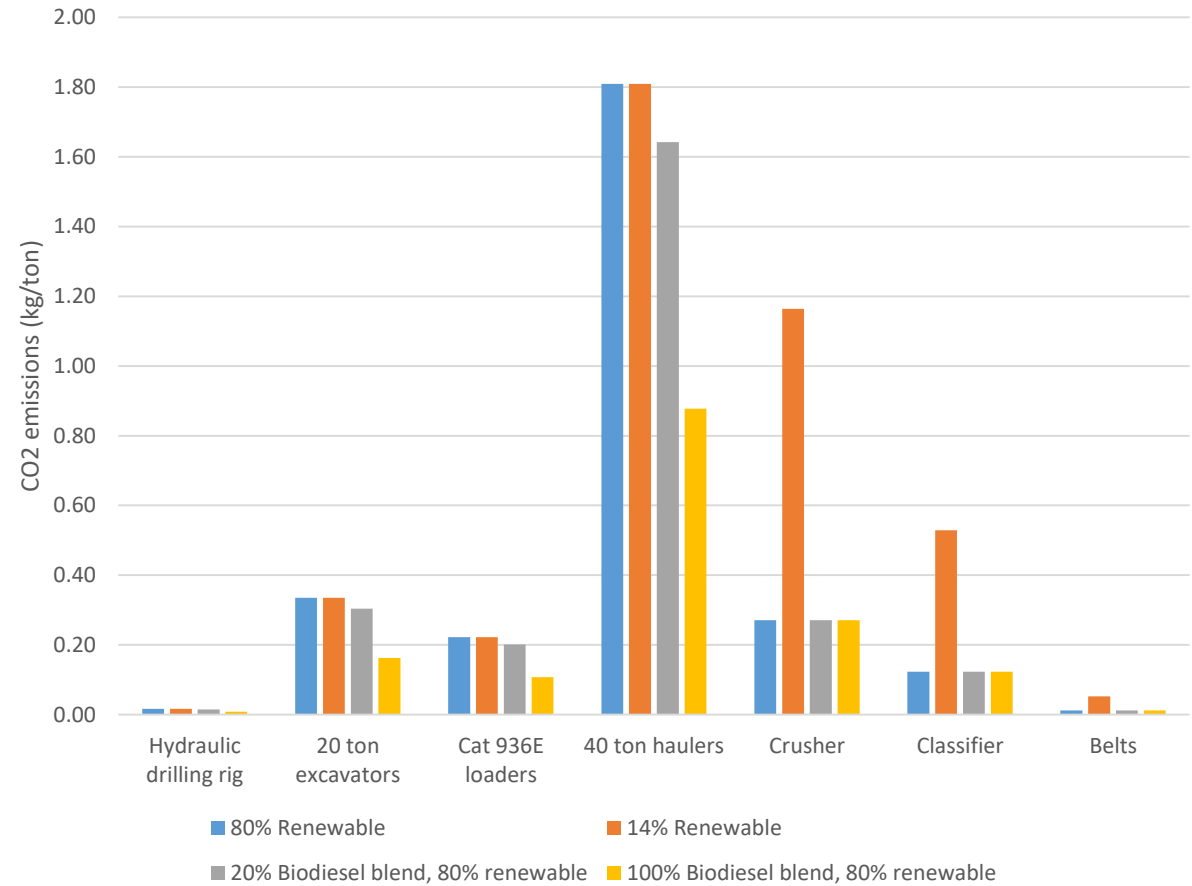
LCA Sensitivity

Fuel type	kg CO2/kg fuel production	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Diesel	0.096	0.0864	0.1056	2.79	2.78	2.8	-0.36	0.36
Biodiesel	-1.53	-1.377	-1.683	2.79	2.79	2.79	0.00	0.00
Methane	0.01	0.009	0.011	2.79	2.79	2.79	0.00	0.00
Coal	0.01	0.009	0.011	2.79	2.79	2.79	0.00	0.00
Fuel type	CV (MJ/kg)	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Methane	52	46.8	57.2	2.79	2.81	2.77	0.72	-0.72
Coal	33	29.7	36.3	2.79	2.82	2.77	1.08	-0.72
Huntly Power Station	Capacity	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Capacity	65	58.5	71.5	2.79	2.75	2.82	-1.43	1.08

What if?

- 20% or 100% biodiesel blend
- 14% renewable electricity

Situation	CO2 emissions (kg/ton)
80% renewable	2.79
14% renewable	4.13
20% biodiesel blend, 80% renewable	2.57
100% biodiesel blend, 80% renewable	1.56



Monte Carlo simulation

- <https://www.youtube.com/watch?v=BQv2Uyea8i4>
- Monte Carlo Simulations in Excel without 3rd Party Add-ins
- 15 variables for the quarry that exist between a range
 - Uniformly distributed
 - Some kind of probability distribution
- Monte Carlo simulation
 - Randomly select those variables
 - Calculate a result
 - Repeat (e.g. up to 10,000 times or more)
 - Gives a probability of a result happening
 - Generate a histogram

Monte Carlo simulation

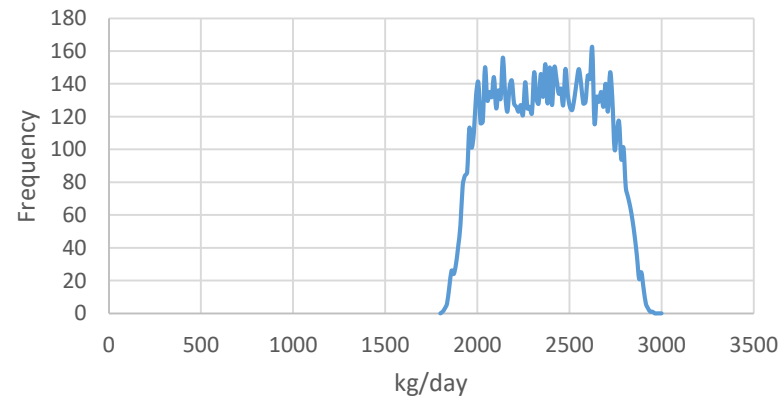
- In Excel use for each variable
 - Randbetween(lower value, upper value) for uniform distribution
 - Norminv(Rand(),mean value, standard deviation) for normal distribution
 - Likes whole positive numbers, not fractions
 - To display results of all calculations:
 - Set up a table with columns simulation# and the output
 - Rows 1:10000
 - Select the whole table
 - Use Data, What-if analysis, Data table
 - Row input cell – leave empty
 - Column input cell – select a cell that is empty and won't have any data
 - Excel will fill up the table with the results from each simulation

Monte Carlo simulation

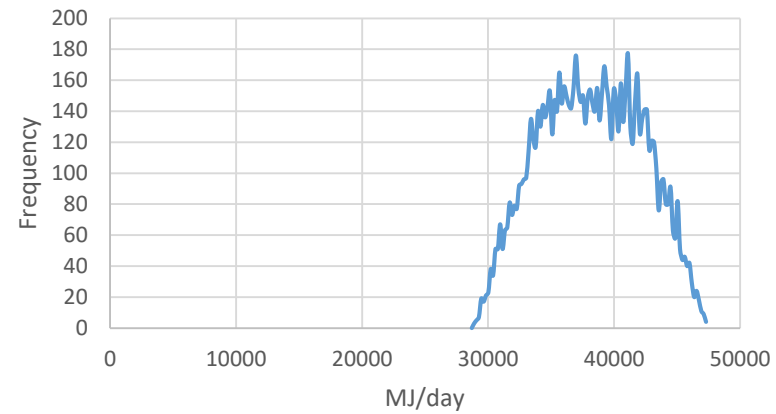
- To generate histograms
 - Turn on the Analysis ToolPak add-in
 - Create a table with the ranges you are collecting frequencies of values between
 - Use Data, Data Analysis, Histogram to generate the Histogram from the simulation results

Monte Carlo Simulation

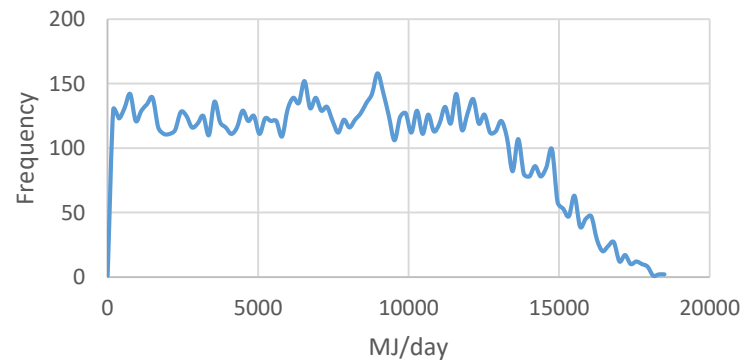
Fuel use



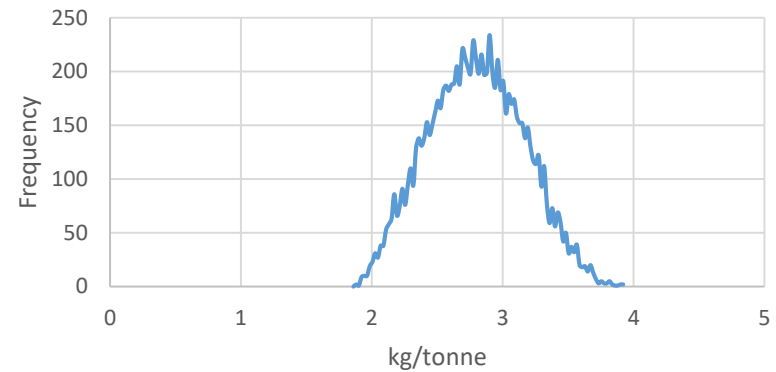
Electricity use



Non renewable electricity use



CO2



Monte Carlo Simulation

	Fuel use (kg)	Electricity (MJ)	Non renewable electricity (MJ)	CO2 emissions (kg/day)	NRE (MJ/ton)	CO2 (kg/ton)	NRE (MJ/\$)	CO2 (kg/\$)
Mean	2362.96	37998.23	7589.86	8896.77	2.37	2.78	0.10	0.12
Median	2367.51	38033.28	7427.04	8888.42	2.32	2.78	0.10	0.12
Min	1801.81	28748.16	0.00	5799.91	0.00	1.81	0.00	0.08
Max	2933.60	47287.30	18479.85	12394.94	5.77	3.87	0.25	0.16
Std. Dev.	264.82	4001.04	4477.03	1172.47	1.40	0.37	0.06	0.02

Data Quality

(Ref: Energy Efficiency and Conservation Authority, Life Cycle Assessment of Electric Vehicles
Final Report, 2015)

- Reliability (actual, estimated, calculated, opinion)
- Completeness (complete – all sites, some sites, one site, unknown)
- Temporal correlation (current, old, really old)
- Geographical correlation (actual, larger or smaller area, different country)
- Technology (data from that under study, related, different scale, lab)
- Sample size (bigger is better)

Pedigree matrix



Score:	1	2	3	4	5
U1 Reliability	Verified data based on measurements	Verified data partly based on assumptions OR non-verified data based on measurements	Non-verified data partly based on qualified estimates	Qualified estimate (e.g. by industrial expert); data derived from theoretical information (stoichiometry, enthalpy, etc.)	Non-qualified estimate
	1.00	1.05	1.10	1.20	1.50
U2 Completeness	Representative data from all sites relevant for the market considered over an adequate period to even out normal fluctuations	Representative data from >50% of the sites relevant for the market considered over an adequate period to even out normal fluctuations	Representative data from only some sites (<<50%) relevant for the market considered OR >50% of sites but from shorter periods	Representative data from only one site relevant for the market considered OR some sites but from shorter periods	Representativeness unknown or data from a small number of sites AND from shorter periods
	1.00	1.02	1.05	1.10	1.20
U3 Temporal correlation	Less than 3 years of difference to our reference year (2000)	Less than 6 years of difference to our reference year (2000)	Less than 10 years of difference to our reference year (2000)	Less than 15 years of difference to our reference year (2000)	Age of data unknown or more than 15 years of difference to our reference year (2000)
	1.00	1.03	1.10	1.20	1.50
U4 Geographical correlation	Data from area under study	Average data from larger area in which the area under study is included	Data from smaller area than area under study, or from similar area		Data from unknown OR distinctly different area (north america instead of middle east, OECD-Europe instead of Russia)
	1.00	1.01	1.02		1.10
U5 Further technological	Data from enterprises, processes and materials under study (i.e. identical technology)		Data on related processes or materials but same technology, OR Data from processes and materials under study but from different technology	Data on related processes or materials but different technology, OR data on laboratory scale processes and same technology	Data on related processes or materials but on laboratory scale of different technology
	1.00		1.20	1.50	2.00
U6 Sample size	>100, continuous measurement, balance of purchased products	>20	> 10, aggregated figure in env. report	>=3	unknown
	1.00	1.02	1.05	1.10	1.20

Data Quality

$$SD_{g95} = \sigma_g^2 \\ = \exp \sqrt{(\ln(U_1))^2 + (\ln(U_2))^2 + (\ln(U_3))^2 + (\ln(U_4))^2 + (\ln(U_5))^2 + (\ln(U_6))^2 + (\ln(U_b))^2}$$

The factors U_1 till U_6 referring to the scores for:

- Reliability (U_1)
- Completeness (U_2)
- Temporal Correlation (U_3)
- Geographical Correlation (U_4)
- Further Technological (U_5)
- Sample Size (U_6)

The factor U_b refers to the basic uncertainty factor and is emission-specific.

Data quality

Data or assumption	Conventional	Electric	Qualitative description (rationale)
New Zealand electricity			For all descriptions relating to electricity below, refer also to Appendix E1.
Coal	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Data from literature-source specific to NZ electricity, described in Section 3.5 and 4.3.
Water	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Black coal	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Assumes density of coal in NZ, based on AusLCI data
Transport (trucking)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Transport (rail)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Emissions to air (carbon dioxide, methane, dinitrogen monoxide, nitrogen oxides, carbon monoxide, NMVOC, sulfur dioxide)	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Calculated, data from literature-source, based on total emissions from electricity generation using coal (2013 MfE data)
Emissions to air (other Australian NPI pollutants)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimated, based on Australian NPI data for electricity from coal
Gas	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Data from literature-source specific to NZ electricity, described in Section 3.5 and 4.3.
Water	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Natural gas	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Assumes carbon content of natural gas in NZ, based on AusLCI data
Transport (trucking)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Transport (rail)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Emissions to air (carbon dioxide, methane, dinitrogen monoxide, nitrogen oxides, carbon monoxide, NMVOC, sulfur dioxide)	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Calculated, data from literature-source, based on total emissions from electricity generation using coal (2013 MfE data)
Emissions to air (Remaining Australian NPI pollutant data)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimated, based on Australian NPI data for electricity from coal
Hydropower	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Literature source
Geothermal	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Literature source

LCA 3 – Setting Specifications

ENGEN370

Mark Lay

Specifications

- Normally an LCA is conducted where the product and method of manufacture is known, and all the quantities of materials, etc are known
- At the moment your projects are largely unconstrained (?)
- Open ended, complex problem, with almost infinite possibilities
- Basically a design problem coupled with an LCA
- Unsolvable



Possibilities



Specifications

- You want to gradually add constraints to start converting a complex problem into one that can be solved
- The first place to start is by setting some specifications for your part of the project: e.g.
- Electricity generation: max capacity (MW), type of generation, efficiency, fuel type, location, cooling, size, number of turbines, plant life time (boilers, turbines, pumps), maintenance periods
- Home heating: heat output (kW), type of heating, size, coefficient of performance, building orientation (passive heating), product life time
- Electric vehicles: type, size, weight, speed, acceleration, capacity, number of wheels, number of motors, vehicle/battery/motor life time

Specifications

- Drinking water: Volume per day, method of distribution, method of treatment, location of plant, raw water quality, drinking water quality, plant life time
- Packaging: purpose, size, weight, thickness, material type (type of glass, plastic, paper, fabric), toughness, biodegradability, single/multiple use
- Housing: floor area, number of levels, foundation type, frame type, cladding, roofing, garage (attached or unattached), number of rooms, number of bathrooms (toilet separate from bathroom), room sizes, general plan, house life time (cladding, roofing, plumbing)
- Bridges: type, size (length, height, width), loading, construction materials, location, bridge life time, maintenance periods

Specifications

- Waste water treatment: capacity, type of treatment, treatment steps, raw water quality (total suspended solids, total dissolved solids, total organic carbon, total nitrogen, total phosphorous, e-coli count, biological oxygen demand, chemical oxygen demand), treated water quality, location, plant life time
- Time keeping: type of watch, size, weight, materials of construction, life time

Specifications – e.g. waste water treatment

Parameter	Before treatment	After treatment	Tunisian standards (N.T.106.002)
TDS (mg/L)	1,350	1,250	–
Salinity (mg/L)	1	1	–
Conductivity (mS/cm)	2.21 ± 0.40^a	2.14 ± 0.15^a	–
Temperature (°C)	35.70 ± 4.78^a	29.36 ± 4.78^a	25
pH	7.25 ± 0.33^a	7.55 ± 0.33^a	$6.5 < \text{pH} < 8.5$
N-NH ₄ ⁺ (mg/L)	39.14 ± 1.98^a	1.65 ± 0.78^a	1
N-NO ₂ ⁻ (mg/L)	4.4 ± 1.88^a	6.43 ± 1.51^a	0.5
N-NO ₃ ⁻ (mg/L)	1.32 ± 0.16^a	13.4 ± 6.26^a	50
BOD ₅ (mgO ₂ /L)	80 ± 4.58^a	26 ± 5.30^a	30
COD (mgO ₂ /L)	146 ± 27.5^a	45.00 ± 28.00^a	90
TSS (mg/L)	40.0 ± 2^a	10.67 ± 2.31^a	30
Fecal coliforms (MPN/100 mL)	5,966.66	1,000	2,000
<i>Faecal streptococci</i> (MPN/100 mL)	110,000	320 ± 34.64	1,000

Comparison of the average wastewater quality before and after treatment to Tunisian standards $\pm$, standard deviations ^a Values are given as a mean of three replications

How to set the specifications?

- Look up what is typical – e.g. Google (existing products, processes, infrastructure), Wikipedia, Branz, Drinking Water NZ Standards, waste water treatment discharge standards, other LCA studies
- E.g. Huntly Power Station:
 - 3x 250 MW Rankine turbines, 30% efficiency, pulverised coal fired boilers, Indonesian coal, 5% ash content, calorific value ~ 30 MJ/kg, water cooled condensers, 50-60 year life time, maintenance on turbine and boiler once every 3 years
 - 1x 400 MW combined cycle (Brayton turbine plus Rankine turbine), 50% efficiency, natural gas, ~ 50 MJ/kg, air cooled, maintenance once every 4 years

How to set the specifications?

- E.g. Tesla model S: Size: 5 m by 2.2 m by 1.44 m, wheelbase 3 m, capacity 5 people, 1020 horsepower, range 412 miles, acceleration 0-60 mph in 2.1 seconds, weight 2.2 tonnes, front and rear AC induction motors
- If unknown, set a specification (until you find information)
- How detailed? Depends on the goal and scope of the LCA and how much time you have, but start off with a high level specification first and gradually add detail as you need it to progress the project: e.g.
 - Huntly – cooling water demand, typical operating capacity
 - Tesla – tires – size, weight, composition

Example specification sheet – e.g. Huntly

Specification	Value	Unit
Location	Huntly	-
Turbine type	Rankine (HP, IP, LP)	-
Number of turbines	3	-
Generation capacity per turbine	250	MW
Boiler type	Suspended, tangential flow, pulverised coal, 1 per turbine	-
Fuel type	Coal - bituminous	-
Fuel source	Kalimantan, Indonesia	-
Coal processing equipment	Grizzly, pulveriser (4 per boiler)	-
Boiler exhaust cleaning equipment	40 kV electrostatic precipitators (2 per boiler)	-
Calorific value of fuel	25	MJ/kg
Fuel composition (ash)	5	%
Boiler and turbine efficiency	33	%
Boiler heat recovery	Superheater, primary, reheater, economiser, air preheater	-
Coal demand (full load) per boiler	30.3	kg/s
Cooling water demand (full load) per turbine (5 °C temperature change)	24.3	m ³ /s
Cooling water supply	Waikato river	-
Air demand (full load) per boiler	337.8	m ³ /s
Typical annual operating capacity	65	%
Building area (m2)	44300	m ²
Site area (m2)	620000	m ²

Activity

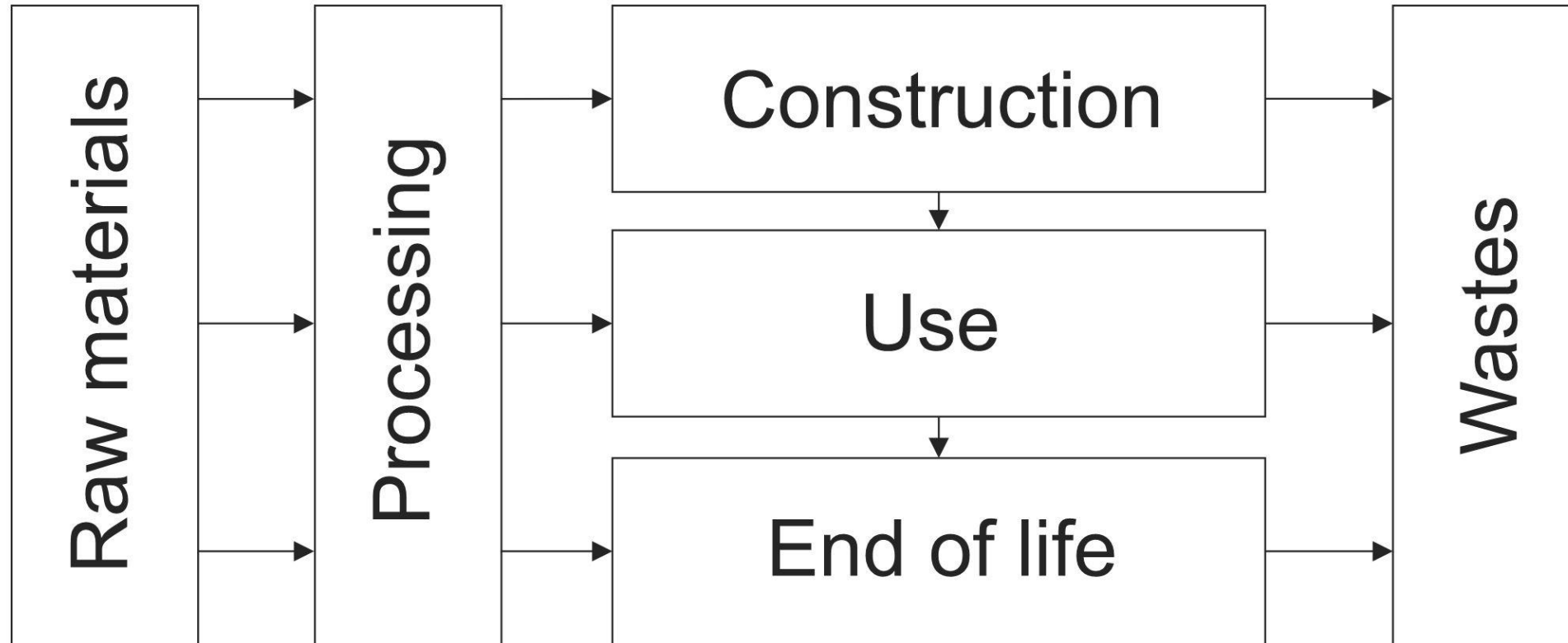
- Using the example specification sheet on Moodle, develop a specification sheet for your product, process, infrastructure

LCA 4 – Developing a life cycle flow chart

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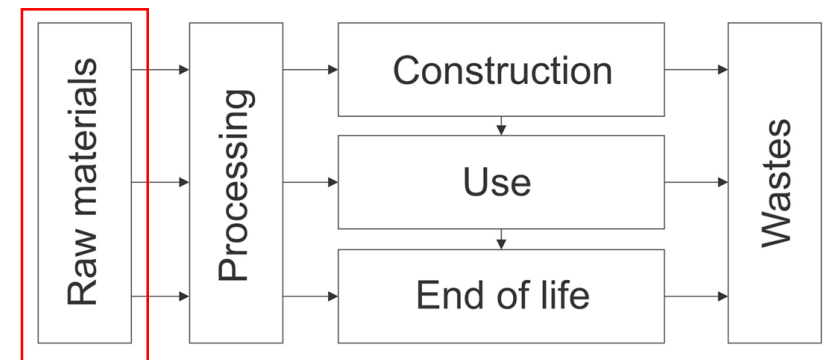
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Common features of LCA flow charts



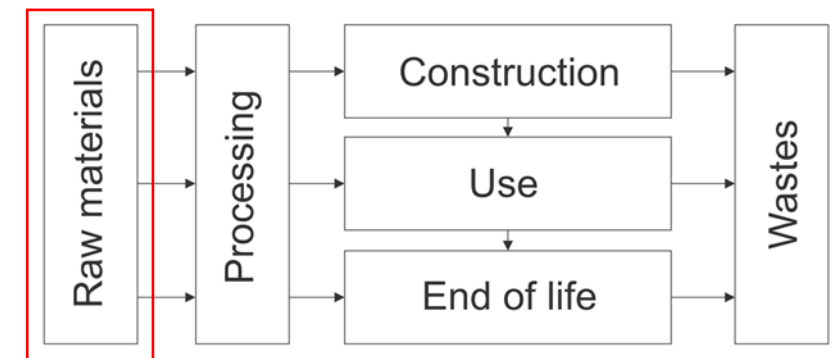
Raw materials

- Raw materials consist of anything that is used to make your product, process or infrastructure, and the extraction processes to recover those materials
- E.g. for concrete – aggregate, gravel, cement (limestone, sand, gypsum), plasticiser, water
- E.g. for a reinforced concrete wall – concrete, steel reinforcing (iron ore, coke), form-work (wood, nails, screws), spacers (plastic)
- E.g. for batteries – lithium salts, plastic, copper
- E.g. for PVC pipes – oil, salt



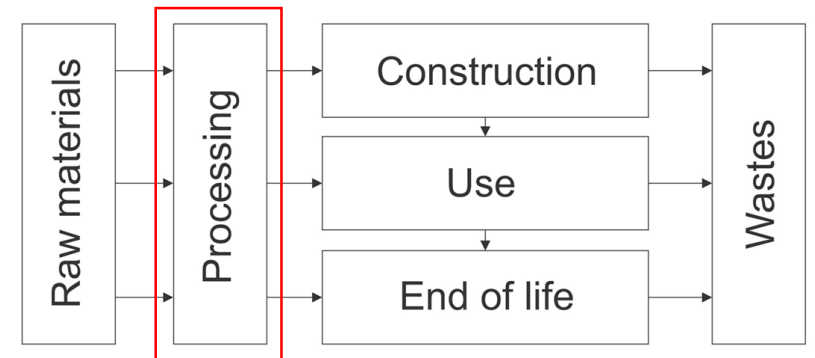
Raw material extraction

- E.g. aggregate, gravel, limestone, sand, gypsum – from a quarry (removing over-burden, digging up the material, crushing, sizing)
- E.g. oil – drilling, or digging if shale oil
- E.g. lithium salts from brine lakes – drilling, pumping, evaporation
- E.g. wood – growing and chopping down trees



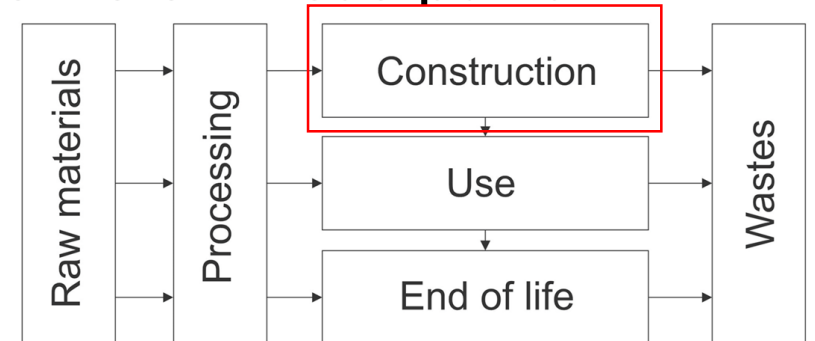
Processing

- Processing involves any step to process the raw materials into a useful form
- E.g. PVC – oil distillation, cracking, refining to produce ethylene, electrolysis of salt to produce chlorine, chlorination of ethylene to produce vinyl chloride, cross linking of vinyl chloride to produce PVC
- Cement – pyrolysis of lime to produce calcium oxide, mixing with gypsum and sand to produce cement



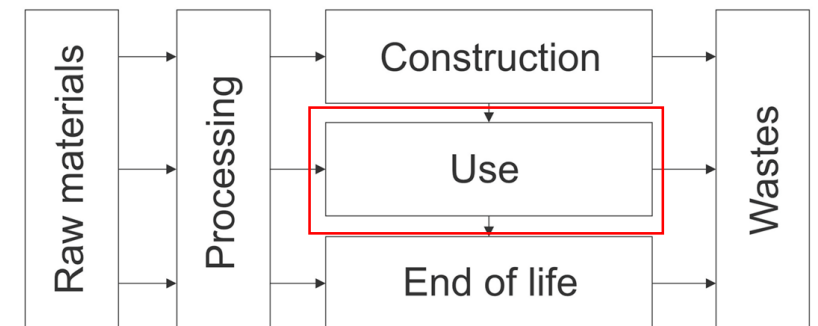
Construction

- This involves any step in the fabrication of your product, plant, or infrastructure
- E.g. a shopping mall: demolition, earth moving, piling, stormwater, water, sewage pipe installation, foundations, steel frame installation, roofing, flooring, cladding, windows, fire suppression, ventilation, heating, air conditioning, lifts, escalators, bathrooms, electric cabling, insulation, internal walls, lights, fittings, painting, carpets (in this case, you might like to include the manufacture of some of these parts under processing)



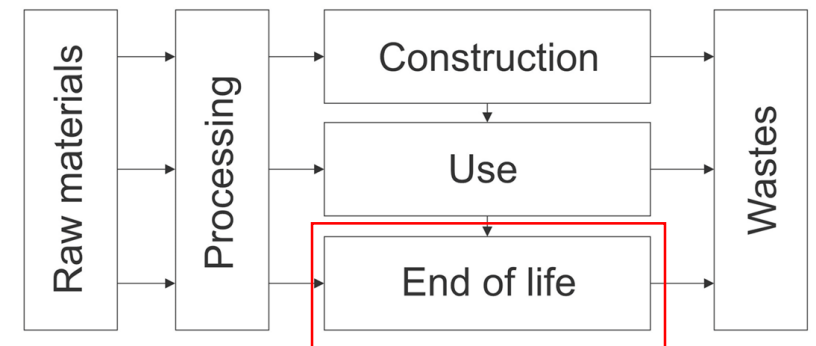
Use

- This involves day to day use of the product, plant or infrastructure, and maintenance
- E.g. car – fuel consumption, oil consumption, tire replacement, coolant use, windscreen wipers, batteries, catalytic converter, replacement of ball joints, push rods, shocks



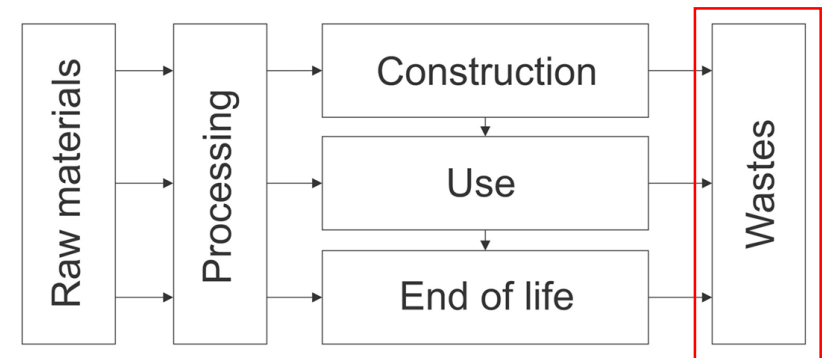
End of life

- This includes any process that occurs once your product, process or infrastructure has reached the end of its useful life
- E.g. shopping mall: demolition, recycling of materials, disposal of materials, rehabilitation
- E.g. car: stripping for parts, crushing, recycling, disposal



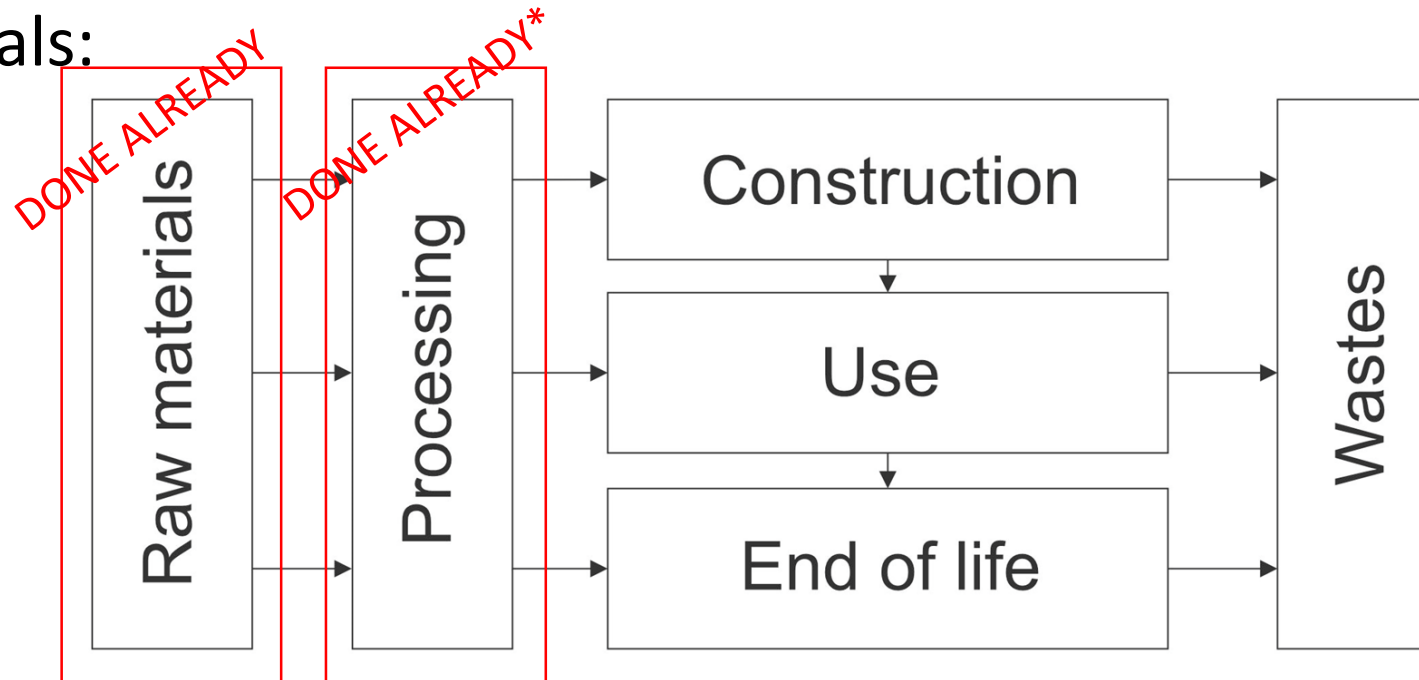
Wastes

- Wastes are anything from the construction, use and demolition of your product, process or infrastructure
- Wastes can include solid waste, atmospheric or water discharges, discharges to land or landfill
- Raw material extraction and processing will generate wastes as well
- Wastes considers what processes are undertaken to treat and discharge the wastes



Developing the flow chart

- In most cases, LCAs around raw material extraction and processing have already been done, so including details about these in the flowchart can be neglected
- Data for these can be found in various LCA databases (e.g. Gabi), and journals:

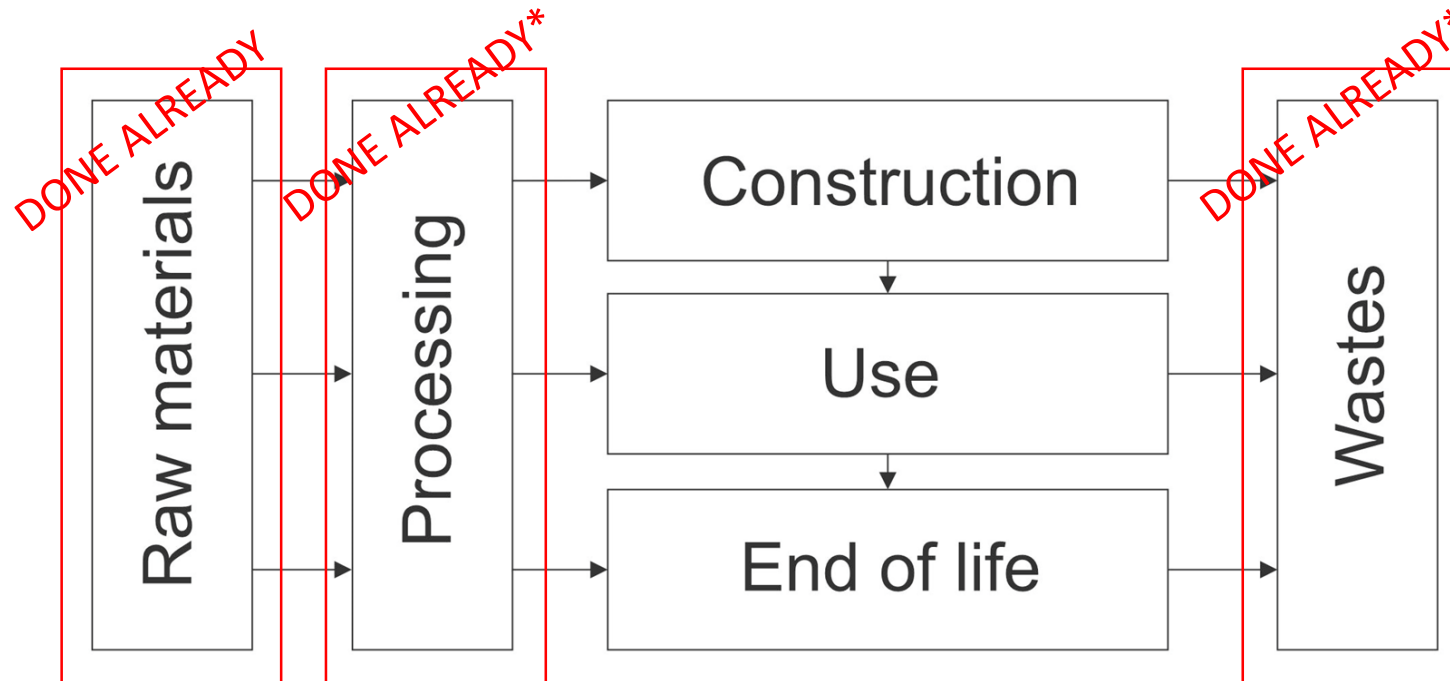


Developing the flow chart

- Should I focus on construction, use and or disposal?
 - This depends on where the focus of the LCA is and the bulk of the environmental impacts
 - E.g. an IC car – bulk of the impact is in the use of the car
 - E.g. an electric car – depends on the country and their electricity mix, the bulk of the impact might be in the fabrication of the car, but in places where electricity is generated by burning fossil fuels, some of the impact will be in the use of the car
 - E.g. a nuclear power plant – the bulk of the impact will be in the construction of the plant and extraction of uranium
 - Unless you are focusing specifically on different options for end of life, you can neglect end of life

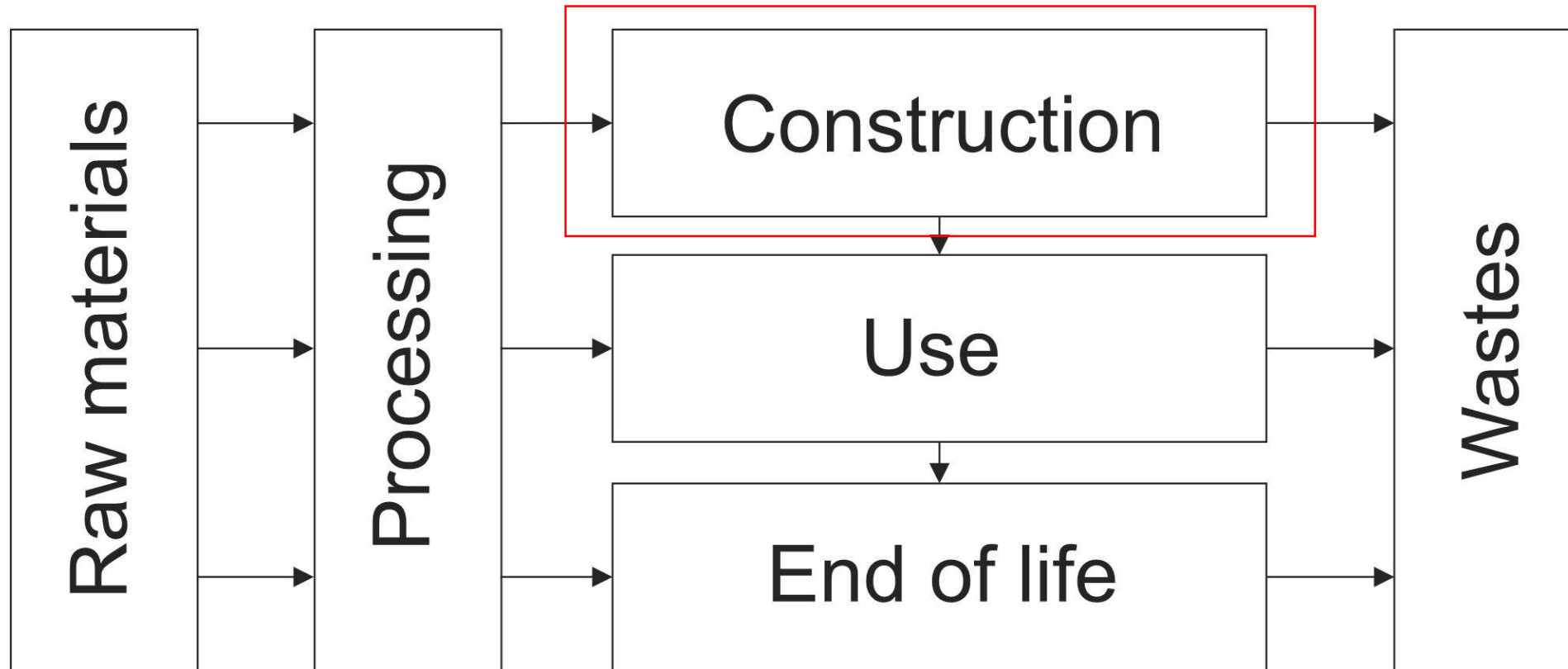
Developing the flow chart

- Should I consider wastes?
 - In many cases, impacts from processing/discharging wastes have already been done

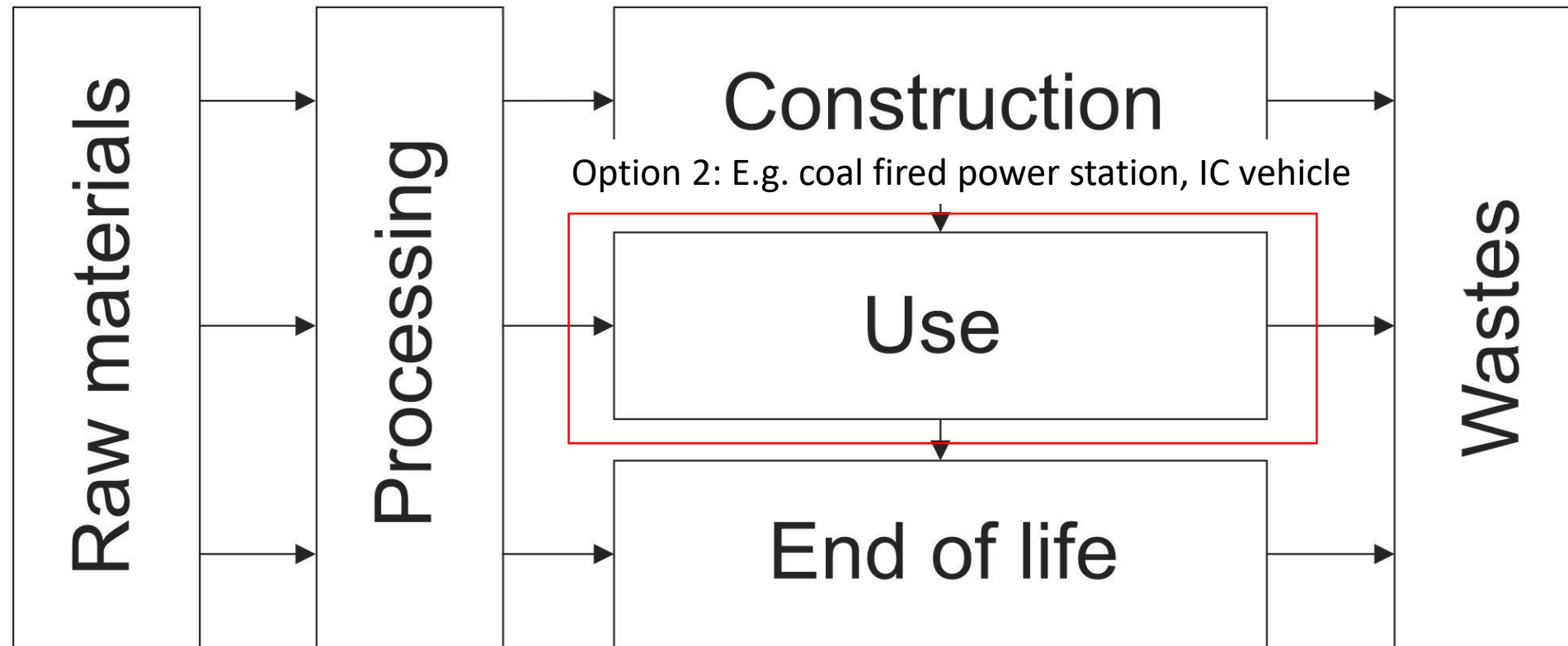


Where to focus efforts?

Option 1: E.g. hydro, solar, nuclear, bridges, may include some processing, will need to decide a life time

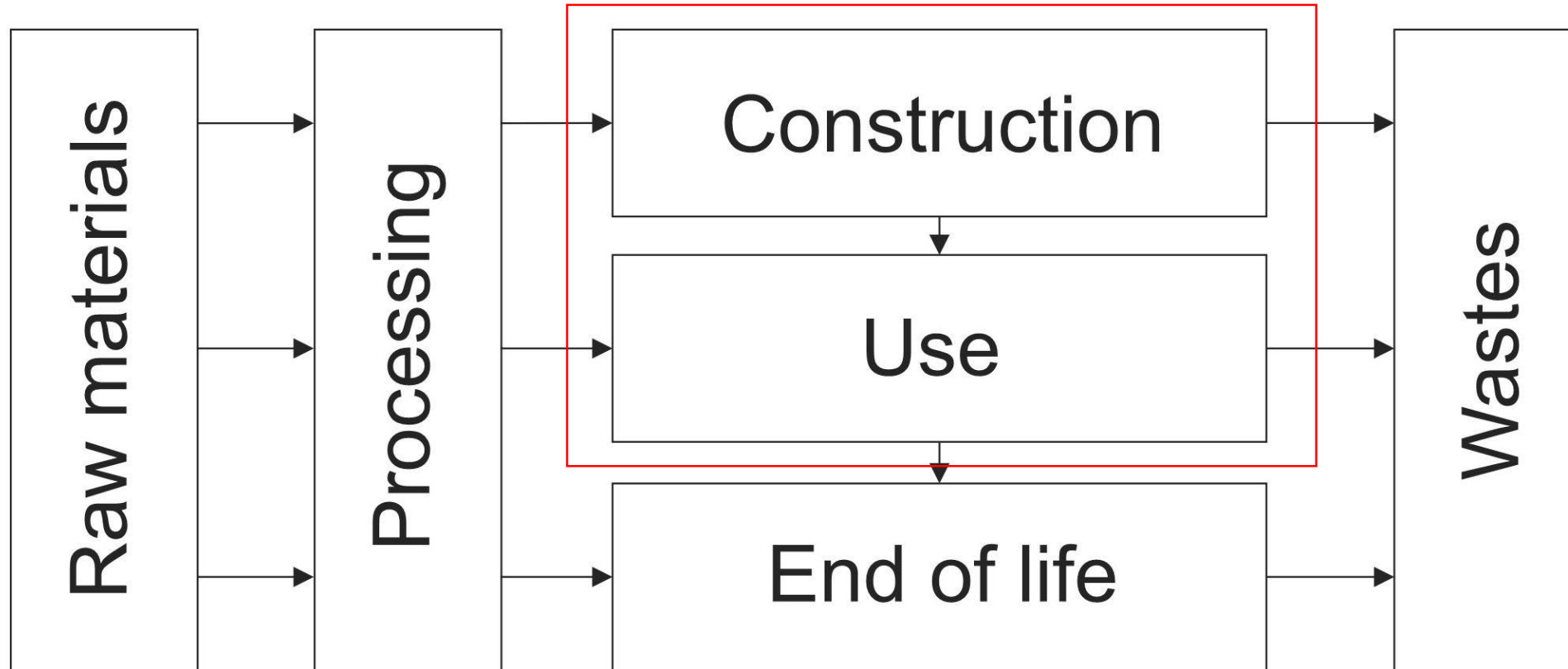


Where to focus efforts?

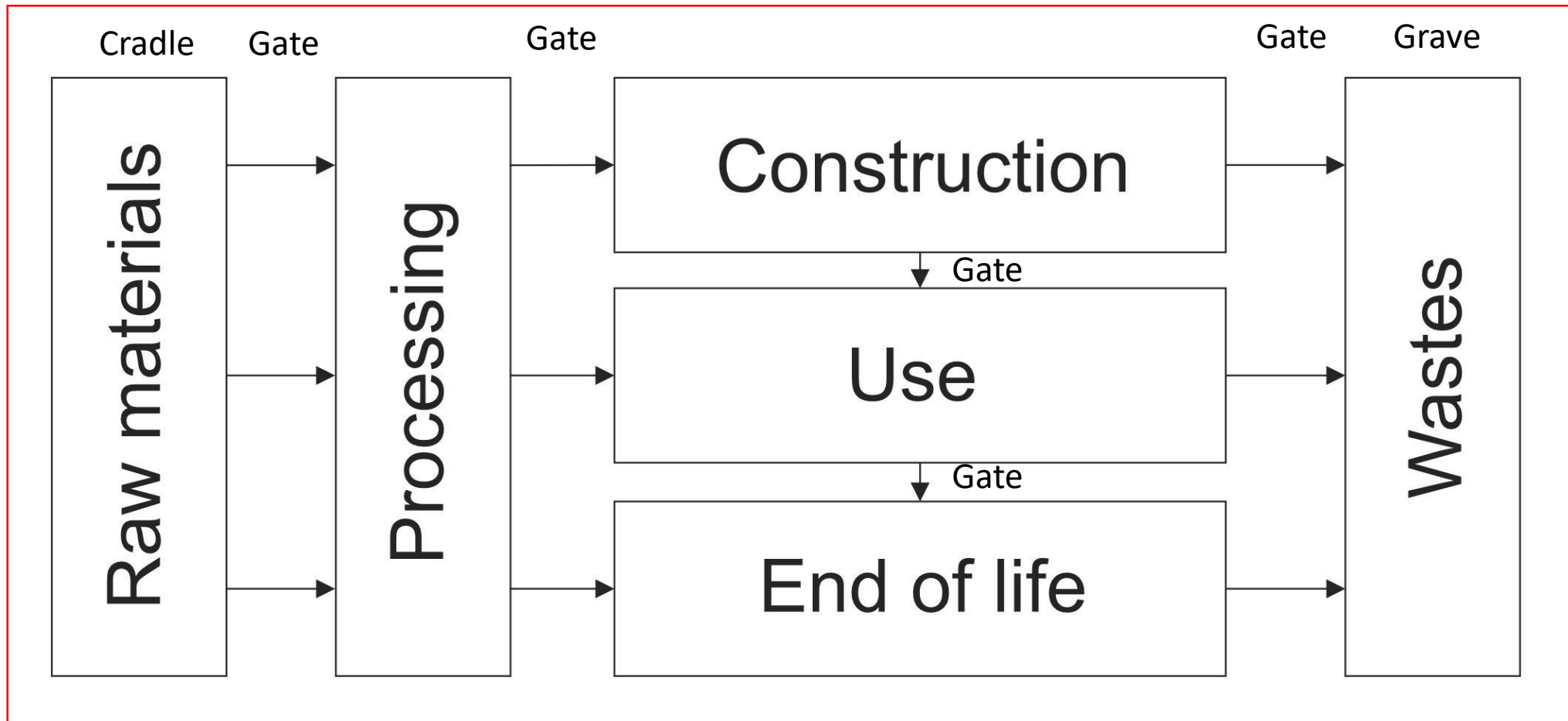


Where to focus efforts?

Option 3: E.g. Passive home heating, electric vehicle, may include some processing, will need to decide a life time

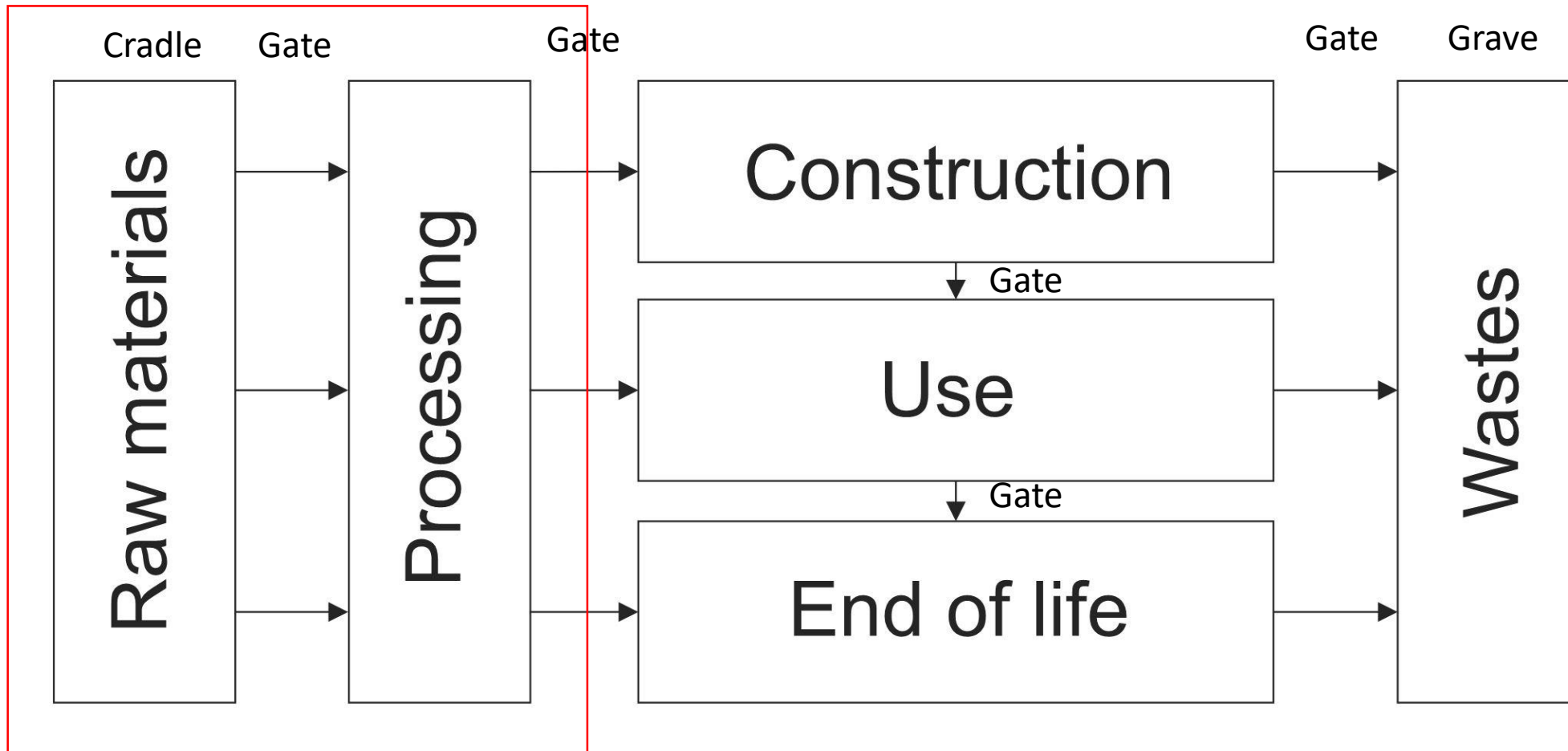


Types of LCA – Cradle to Grave



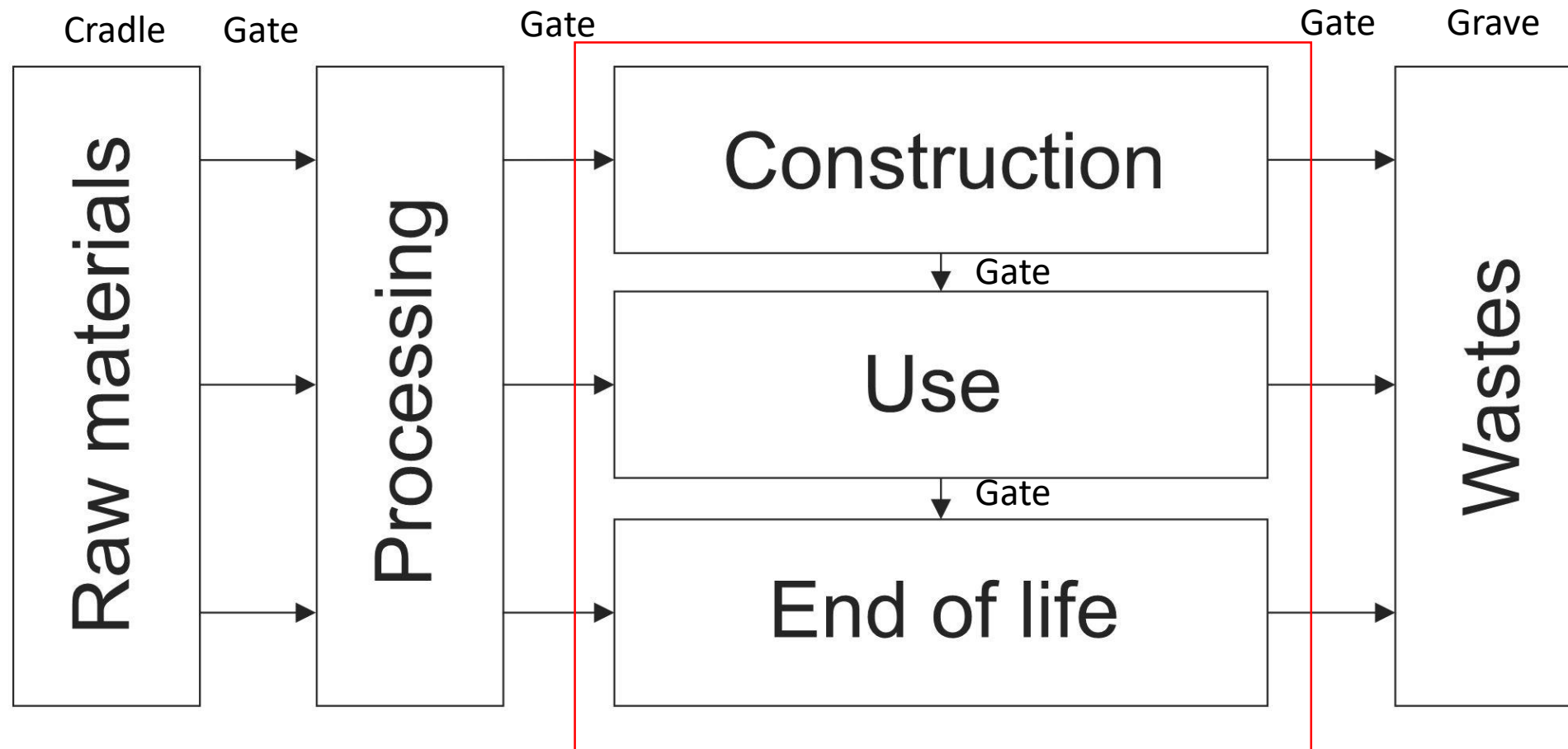
Types of LCA – Cradle to Gate

Can be up to any of the gates



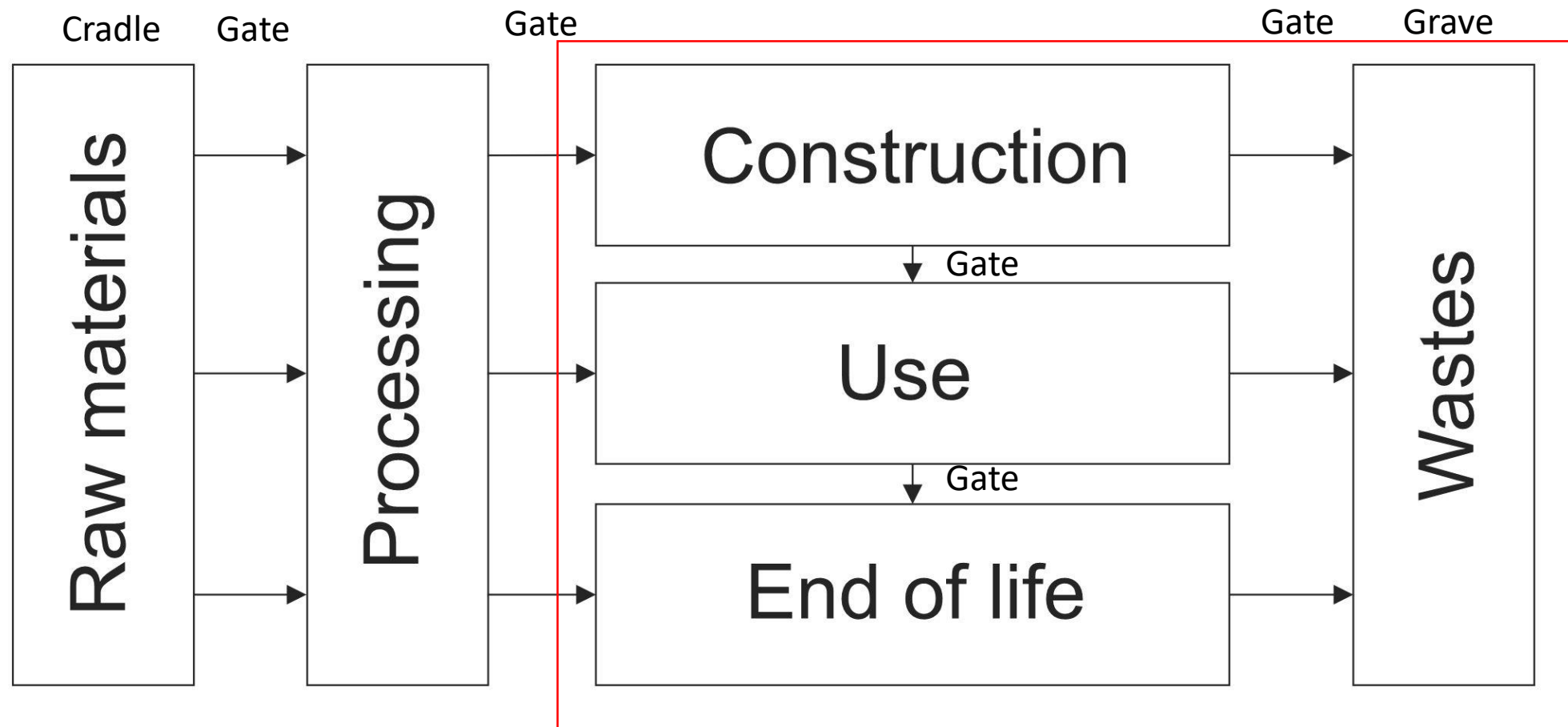
Types of LCA – Gate to Gate

Can be between any of the gates



Types of LCA – Gate to Grave

Can be from any of the gates



Developing a detailed flow chart

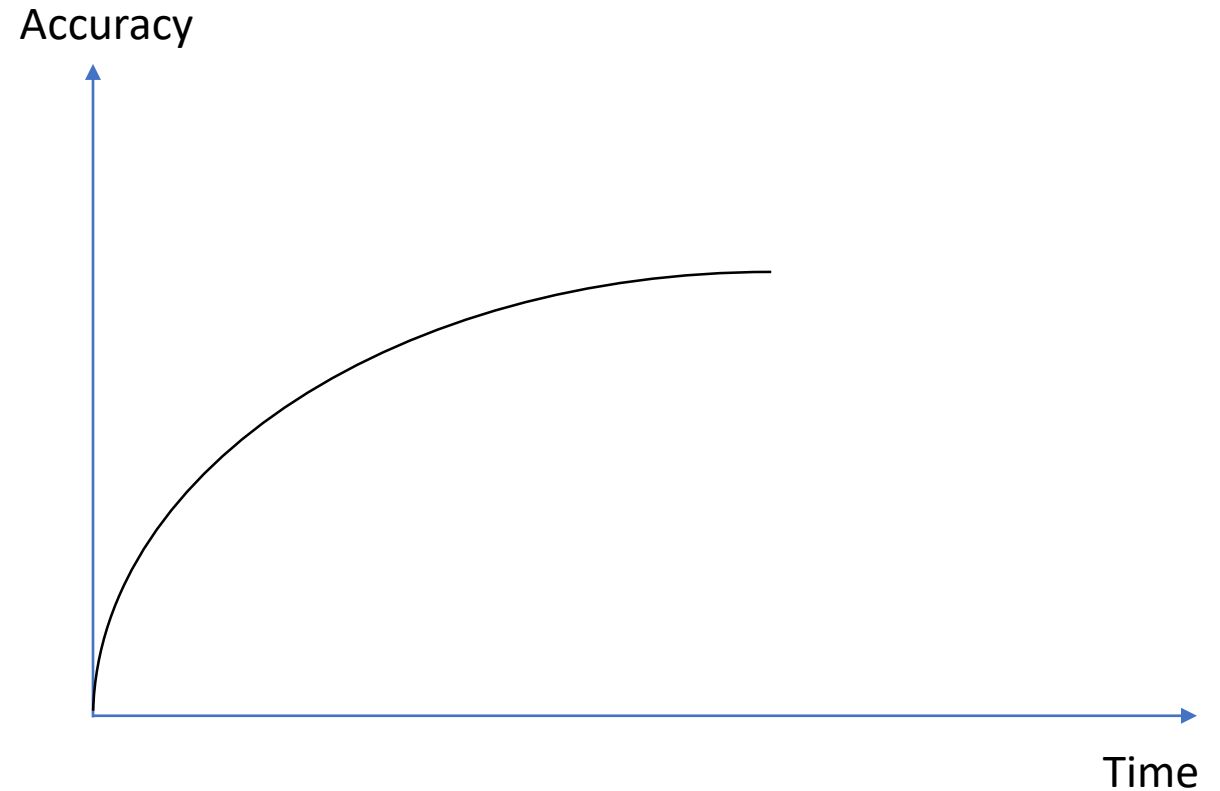
- Once the focus has been decided, you can provide more detail for the construction and/or use
- E.g. a watch:
 - Cutting billet (e.g. if aluminium) for the casing
 - Machining the billet to make the case (e.g. CNC machining)
 - Circuit board manufacture (might like to break this down into steps)
 - Production of motors, gears, dials, glass, buttons
 - Production of batteries
 - Assembly
 - Testing
 - Packaging

Developing a detailed flow chart

- E.g. a coal fired power plant
 - Transporting coal
 - Sizing coal
 - Pulverising coal
 - Blowing coal into the boiler
 - Blowing air into the boiler
 - Burning, ash collection
 - Heat recovery (super heaters, primary, reheater, economiser, air preheater)
 - Exhaust cleaning (electrostatic precipitator)
 - Maintenance (e.g. daily, weekly, monthly, yearly...)

Developing a detailed flow chart

- Where to get information?
 - Google
 - Wikipedia
 - Books (Library)
 - LCA reports
 - Manufacturers websites
 - Company data
 - Physically going there



Activity

- Draw a general flow diagram for your process, product, infrastructure
- Decide on the focus (scope) of your LCA
- Determine the construction and/or use steps (and any processing steps if needed) for your part of the LCA
- Expand the flow diagram to include those steps
- Consider what the material flows in and out might be for each step and add those in (you don't need quantities)
- What type of LCA is your part of your group LCA?

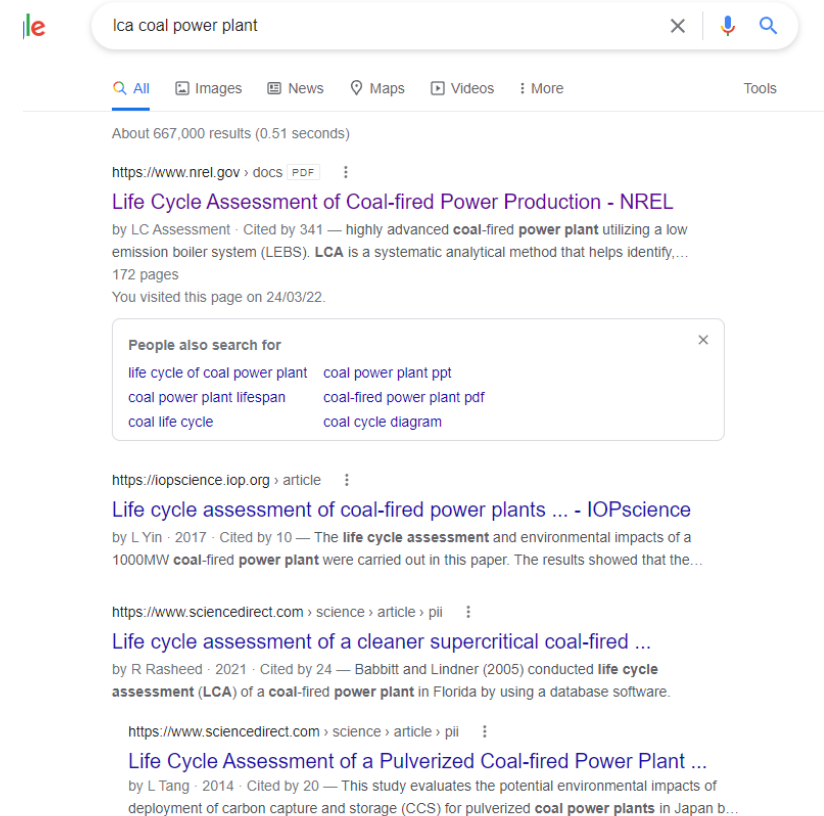
LCA 5 – Literature Review

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Literature Review Tips

- Google search – “LCA coal power plant”
- Pick the one with the most citations first
- Skim through and look at the tables and graphs
- Look through the references (working backwards)
- Look who cited the paper (working forward)
- Don’t spend too much time, probably 5 references is enough to get started
- Have enough information from 2 references to do a basic LCA around a coal fired power plant of any size (assuming linear scaling)



Spath, Mann, and Kerr 1999, Life Cycle Assessment of Coal-fired Power Production

Materials for construction

Table 42: Plant Material Requirements Used in Sensitivity Analysis

Material	Base case (kg/MW plant capacity)	Sensitivity - case D (kg/MW plant capacity)
Concrete	158,758	6,962
Steel	50,721	1,764
Aluminum	419	310
Iron	619	622

Resources used

Table 37: Resource Consumption

Resource	Average (a, b)		NSPS (a, b)		LEBS (a, b)	
	% by wt (c)	g/kWh (d)	% by wt (c)	g/kWh (d)	% by wt (c)	g/kWh (d)
Coal	80.4	475.62	78.0	434.92	97.3	353.37
Limestone	17.4	102.84	19.7	109.49	0.0	0.04
Oil	1.9	11.48	2.0	11.32	1.3	4.88
Natural gas	0.2	1.25	0.2	1.26	1.3	4.53
Iron ore	0.0	0.11	0.0	0.11	0.0	0.095
Iron scrap	0.0	0.12	0.0	0.12	0.0	0.10

- (a) Results are reported for the surface mining case. However, the underground mining numbers are listed here.
- (b) Coal transportation = average user by river.
- (c) Percent of total resource consumption. Not all resources consumed by the system are shown; they do not add up to 100%.
- (d) Resource consumption per kWh of net electricity produced averaged over the life of the system.

Emissions

Table 29: CO₂ Results

System	Total CO ₂ emissions (g/kWh) (a)	CO ₂ from surface mining subsystem (b)		CO ₂ from transportation subsystem (c)		CO ₂ from power generation subsystem	
		(g/kWh) (a)	(%)	(g/kWh) (a)	(%)	(g/kWh) (a)	(%)
Average	1,022	9	0.9	17	1.7	996	97.3
NSPS	941	8	0.9	16	1.7	917	97.4
LEBS	741	7	1.0	13	1.8	721	97.3

- (a) CO₂ emissions per kWh of net electricity produced averaged over the life of the system.
- (b) Results are reported for the surface mining case. However, the underground mining numbers are similar to those listed here.
- (c) Coal transportation = average user by river.

Wastes

Table 54: Power Plant FGC Waste and Ash

Solid waste	Emission amount (g/kWh) (a)		
	Average	NSPS	LEBS
FGC waste - dry total	86.0 (b)	91.5 (b)	34.2 (c)
FGC waste - dry landfilled	64.7	68.9	0
Ash - moisture free total	35.7 (d)	32.6 (d)	26.6 (d)
Ash - moisture free landfilled	25.7	23.5	19.1
Total waste produced from the system (mining, transportation, and power generation subsystems) (e), (f), (g)	110.4	113.4	20.7

Yin, Liao, Zhou, Wang, and Ma 2017, Life cycle assessment of coal-fired power plants and sensitivity analysis of CO2 emissions from power generation side

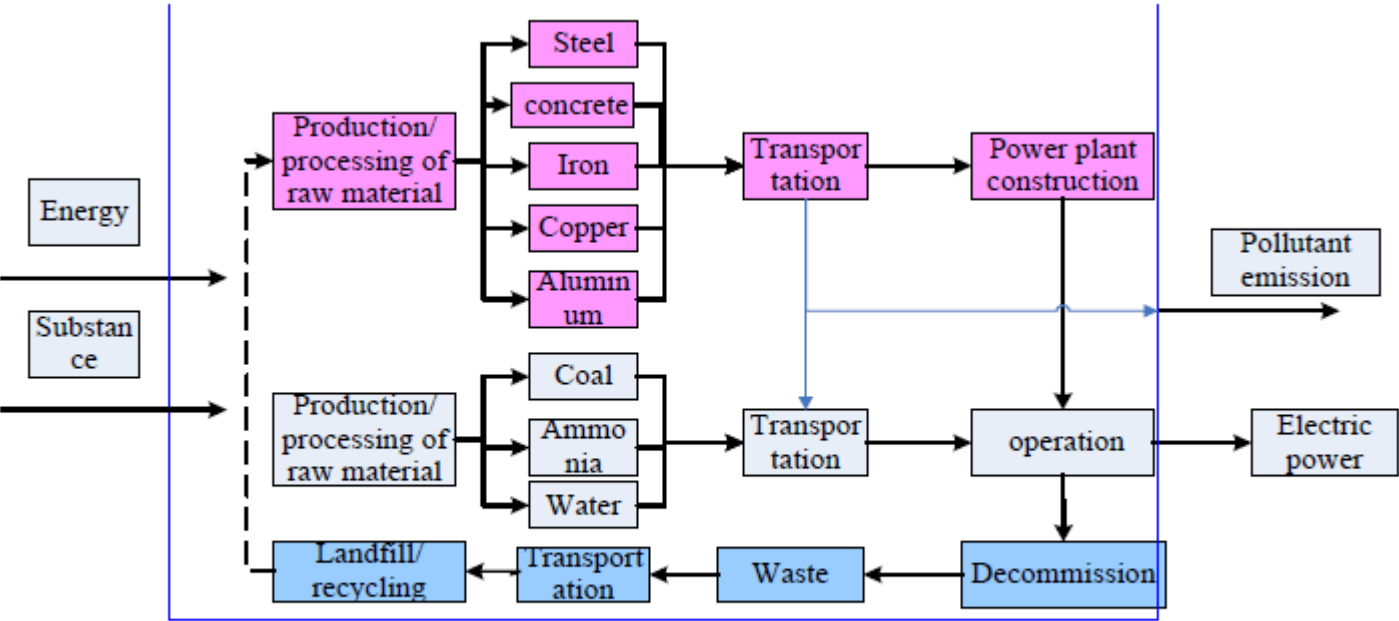


Figure 1. System boundary

Funny units: energy consumption (kg coal/tonne material processed and transported)

Look for conversions: 6.52 kWh = 2.1842 kg coal

Energy consumption and emissions for mining, smelting, and transporting raw materials in China					
Conversion	2.1842	kg coal	=	6.52	kWh
		Energy consumed (kg coal/tonne)		Energy (kWh/tonne)	CO2 (kg/tonne)
		Material			
		Steel	1524	4548	2436
		Concrete	192	572	689
		Fe	536	1599	1536
		Cu	120	359	18382
		Al	78	232	822

Table 1. The energy consumption required for mining, smelting and transporting of raw materials and gaseous pollutants (kg/t)

Item	Standard coal	CO ₂	SO ₂	NO _x	CH ₄	CO	VOC	N ₂ O	COD	Dust
Steel	1523.62	2435.862	5.914794	5.242415	0.009022	0.80976	0.178762	0.004041	0.254	93.42482
Concrete	191.6065	689.364	0.813211	1.37852	0.896079	0.12581	—	—	0.242358	1.031
Fe	535.73	1535.862	4.034742	5.242415	0.009022164	0.809759883	0.178762	0.00404137	—	32.72135
Cu	120.1	18381.8	1298.1	54.1						
Al	77.8	822	2.09	2.98	9.16	0.761	—	—	—	2.39

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Life cycle assessment of coal-fired power production

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[BOOK] Environmental life cycle assessment of goods and services: an input-output approach

CT Hendrickson, LB Lave, HS Matthews - 2010 - taylorfrancis.com

Environmental life cycle assessment is often thought of as cradle to grave and therefore as the most complete accounting of the environmental costs and benefits of a product or ...

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[HTML] Life cycle energy and environmental performance of a new university building: modeling challenges and design implications

C Scheuer, GA Keoleian, P Reppe - Energy and buildings, 2003 - Elsevier

A comprehensive case study life cycle assessment (LCA) was conducted of a 7300 m², six-story building with a projected 75 year life span, located on the University of Michigan ...

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Full cost accounting for the life cycle of coal

PR Epstein, JJ Buonocore, K Eckerle... - Annals of the New York Academy of Sciences, 2011 - Wiley Online Library

Each stage in the life cycle of coal—extraction, transport, processing, and combustion—generates a waste stream and carries multiple hazards for health and the environment ...

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[PDF] wiley.com

Full View

[HTML] Environmental impact and life cycle assessment (LCA) of traditional and 'green'concretes: Literature review and theoretical calculations

P Van den Heede, N De Belie - Cement and Concrete Composites, 2012 - Elsevier

With the current focus on sustainability, it is necessary to evaluate concrete's environmental impact properly, especially when developing new 'green'concrete types. Therefore, we ...

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Life cycle assessment of hydrogen production via natural gas steam reforming

PL Spath, MK Mann - 2000 - osti.gov

A life cycle assessment of hydrogen production via natural gas steam reforming was performed to examine the net emissions of greenhouse gases as well as other major ...

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Renewable energy from willow biomass crops: life cycle energy, environmental and economic performance

GA Keoleian, TA Volk - BPTS, 2005 - Taylor & Francis

[PDF] tandfonline.com

Full View

LCA around concretes

LCA around biomass

Evaluating the information

- Context – is it for a similar or same application?
- Location – is it for a similar or different geographical location?
 - E.g. Yin 2017 paper, Table 1 is for materials made in China – where are you getting your materials from?
 - Metals – probably China
 - Concrete – New Zealand
 - Aluminium – New Zealand – but bauxite mined in Australia
 - Is it a similar process for manufacture?
 - Aluminium made in NZ will have a lower environmental impact
 - Concrete manufacture will depend on the kiln and fuel used for making the calcium oxide from limestone
 - Emissions from energy use will depend on the country and its energy mix

Evaluating the information

- Time – is the data from within the last 5-10 years?
 - Has the method of manufacture changed
 - Has there been any improvements in efficiency of manufacture
- Method of calculating the data
 - Actual measurements (high quality)
 - Based on databases (good/high quality)
 - Based on other journal papers (ok/good quality)
 - Based on calculations (ok quality)
 - Based on assumptions (low quality)
 - Based on opinion, back of the packet calculations (very low quality)

Reporting the paper in your LCA

- Keep it concise
 - E.g. Spath, Mann, and Kerr (1999) carried out an LCA around three types of coal fired power generation and found that 97% of the CO₂ emissions was from electricity generation over the lifetime of the plants.
- Include any key tables/figures from the paper and reference those tables/figures in your text
 - E.g. Spath, Mann, and Kerr (1999) found that concrete was the main material used in the construction of coal fired power plants (Table 1 or whatever table number it is).
- Include some critique if needed
 - E.g. Spath, Mann, and Kerr (1999)'s LCA was based on measured plant data from the 1990's, therefore the data may be out of date.

Activity

- Use Google to find 2 or more papers related to your LCA
- Choose one and find a relevant table that will be useful for your LCA
- Looking through the references, find a paper that will be relevant to your LCA
- Choose one and use Google Scholar to find who has cited your paper, and pick one that is relevant to your LCA
- Choose one parameter from the paper and evaluate the quality of that parameter

LCA 6 – Selecting Impact Category, Function and Functional Unit

ENGEN370

Mark Lay

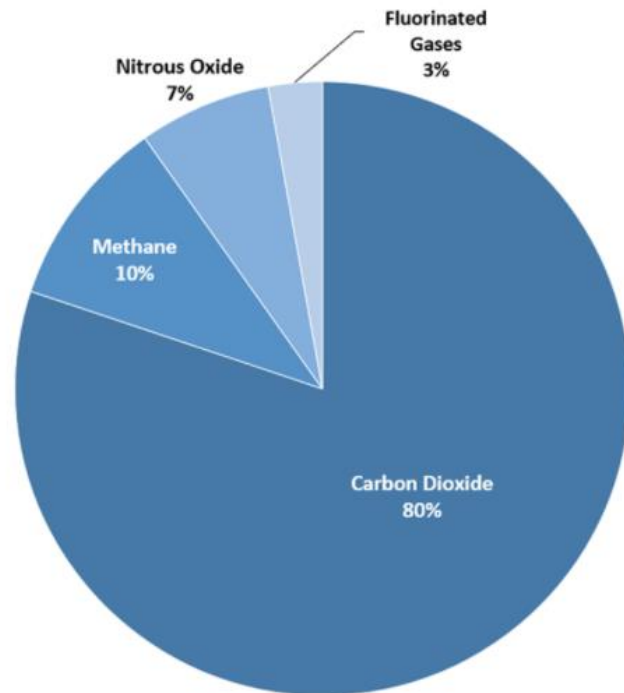
Selecting environmental impacts

<https://ecochain.com/knowledge/impact-categories-lca/>

The ISO Standard requires that a study selects multiple impacts that encompass
“a comprehensive set of environmental issues”

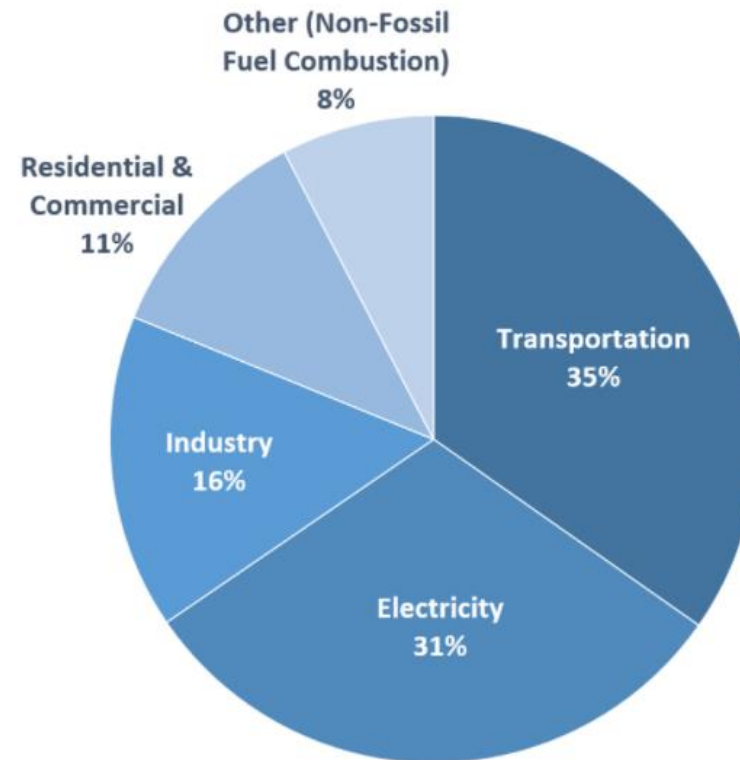
Impact category / Indicator	Unit	Description
Climate change – total, fossil, biogenic and land use (compulsory)	kg CO ₂ -eq	Indicator of potential global warming due to emissions of greenhouse gases to air. Divided into 3 subcategories based on the emission source: (1) fossil resources, (2) bio-based resources and (3) land use change. Calculated by kg of GHG emitted multiplied by a factor to give kg CO ₂ -eq
Ozone depletion	kg CFC-11-eq	Indicator of emissions to air that cause the destruction of the stratospheric ozone layer
Acidification	kg mol H ⁺	Indicator of the potential acidification of soils and water due to the release of gases such as nitrogen oxides and sulphur oxides
Eutrophication – freshwater	kg PO ₄ -eq	Indicator of the enrichment of the fresh water ecosystem with nutritional elements, due to the emission of nitrogen or phosphor containing compounds
Eutrophication – marine	kg N-eq	Indicator of the enrichment of the marine ecosystem with nutritional elements, due to the emission of nitrogen containing compounds.
Eutrophication – terrestrial	mol N-eq	Indicator of the enrichment of the terrestrial ecosystem with nutritional elements, due to the emission of nitrogen containing compounds.
Photochemical ozone formation	kg NMVOC-eq	Indicator of emissions of gases that affect the creation of photochemical ozone in the lower atmosphere (smog) catalysed by sunlight.

Overview of U.S. Greenhouse Gas Emissions in 2019



Total U.S. Emissions in 2019 = 6,558 [Million Metric Tons of CO₂ equivalent](#) (excludes land sector). Percentages may not add up to 100% due to independent rounding.

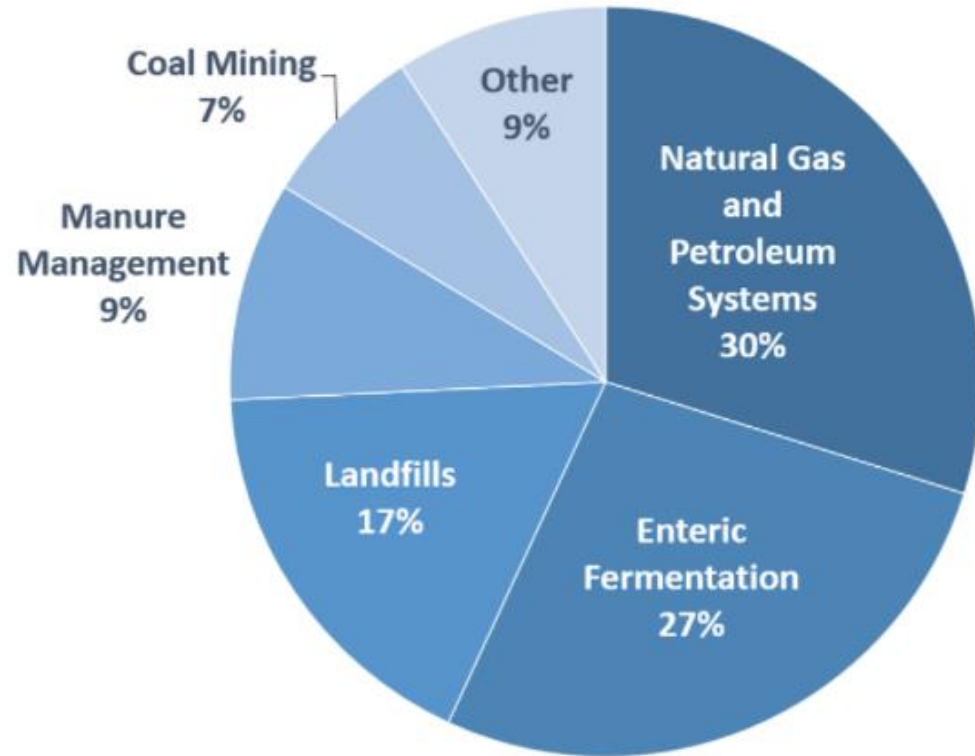
2019 U.S. Carbon Dioxide Emissions, By Source



Note: All emission estimates from the [Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019](#) (excludes land sector).

<https://www.epa.gov/ghgemissions/overview-greenhouse-gases#overview>

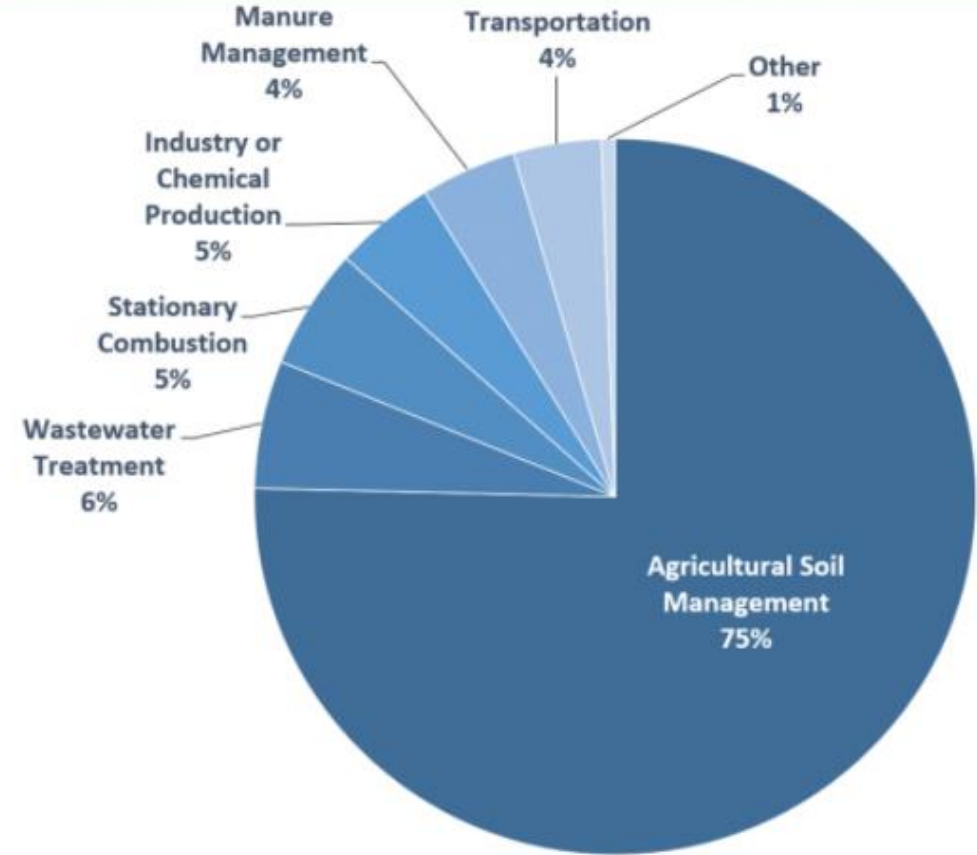
2019 U.S. Methane Emissions, By Source



Note: All emission estimates from the [*Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019*](#) (excludes land sector).

GWP = 25x CO₂

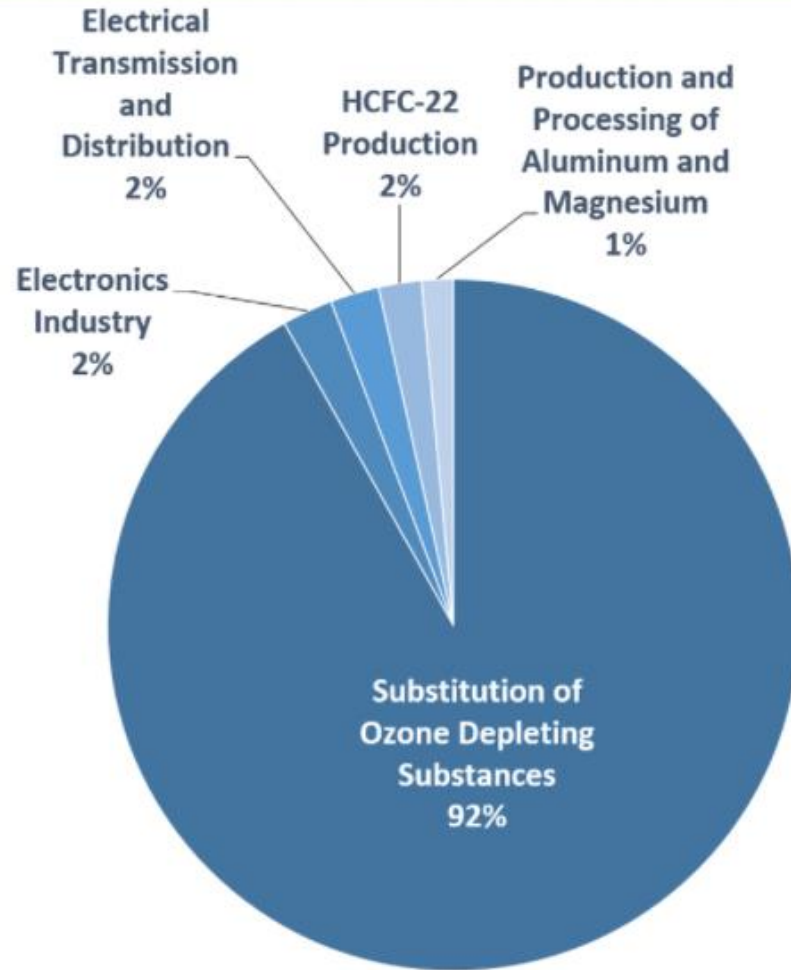
2019 U.S. Nitrous Oxide Emissions, By Source



Note: All emission estimates from the [*Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019*](#) (excludes land sector).

GWP = 298x CO₂

2019 U.S. Fluorinated Gas Emissions, By Source



Note: All emission estimates from the [Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019](#).

Properties of F-gases

Chemical Formulas:

HFCs, PFCs, NF_3 , SF_6

Lifetime in Atmosphere:

HFCs: up to 270 years

PFCs: 2,600–50,000 years

NF_3 : 740 years

SF_6 : 3,200 years

Global Warming

Potential (100-year):¹

HFCs: up to 14,800

PFCs: up to 12,200

NF_3 : 17,200

SF_6 : 22,800

Selecting environmental impacts

- Data available for calculating GHG emissions (in kg CO₂-eq)
 - <https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2018>
 - Ministry for the Environment. 2020. Measuring Emissions: A Guide for Organisations: 2020 Detailed Guide. Wellington: Ministry for the Environment

Selecting environmental impacts

Impact category / Indicator	Unit	Description
Depletion of abiotic resources – minerals and metals	kg Sb-eq	Indicator of the depletion of natural non-fossil resources. Normalised to Antimony (Sb) by multiplying individual materials by a conversion factor. If only a few abiotic resources you can use kg [material] used.
Depletion of abiotic resources – fossil fuels	MJ, net calorific value	Indicator of the depletion of natural fossil fuel resources. Mass of fossil fuel used (kg) multiplied by the calorific value of the fuel (MJ/kg)
Human toxicity – cancer, non-cancer	CTUh	Impact on humans of toxic substances emitted to the environment. Divided into non-cancer and cancer related toxic substances.
Eco-toxicity (freshwater)	CTUe	Impact on freshwater organisms of toxic substances emitted to the environment.
Water use	m3 world eq. deprived	Indicator of the relative amount of water used, based on regionalized water scarcity factors.
Land use	Dimensionless	Measure of the changes in soil quality (Biotic production, Erosion resistance, Mechanical filtration).
Ionising radiation, human health	kBq U-235	Damage to human health and ecosystems linked to the emissions of radionuclides.
Particulate matter emissions	Disease incidence	Indicator of the potential incidence of disease due to particulate matter emissions.

Selecting environmental impacts

Parameter	Unit	Description
Primary renewable energy (materials)	MJ	Use of renewable primary energy resources as raw materials
Primary renewable energy (energy)	MJ	Use of renewable primary energy, excluding renewable primary energy resources used as energy
Primary renewable energy (total)	MJ	Sum of the two values above
Primary non-renewable energy (materials)	MJ	Use of non-renewable primary energy resources as raw materials
Primary non-renewable energy (energy)	MJ	Use of non-renewable primary energy, excluding renewable primary energy resources used as energy
Primary non-renewable energy (total)	MJ	Sum of the two values above
Use of secondary material	kg	Material recovered from previous use or from waste which substitutes primary materials
Use of fresh water	m3	Freshwater use in absolute values
Use of renewable secondary fuels	MJ	Renewable fuel recovered from previous use or from waste which substitutes primary fuels
Use of non-renewable secondary fuels	MJ	Non-renewable fuel recovered from previous use or from waste which substitutes primary fuels

Deciding on the function of your product/process or infrastructure

- In most cases it is obvious
 - Power plant – produce electricity to supply a country or location
 - Housing – to provide warmth/weather protection for people, store belongings....
 - Watch – to tell you the time
 - Clothing – to provide warmth/weather protection/privacy
 - Vehicles – to transport people/goods from place to place
 - Roading – to allow vehicles to travel from place to place

Deciding on the functional unit

- Depends on the product/process or infrastructure
 - Power plant – e.g. per MJ, kWhr, MWhr, MW
 - Housing – e.g. per house, floor area (m²), or person sheltered
 - Watch – e.g. per item
 - Clothing – e.g. per item (standardised to a particular size)
 - Vehicles – per vehicle, person or kg goods carried per km
 - Roading – per length (e.g. km), capacity (vehicles/day in one direction)
 - Heating – per kW heat output
- Look through other LCAs done in your area to see what they used

Types of LCA

- Attributional – Looking at the impact of one system only (what we are doing)
- Consequential – Comparing effects of decisions made (what we are doing)
 - Choosing between two or three methods of asphalt manufacture
 - Choosing between using recycled concrete or new concrete
- Process LCA – (what we are doing)
- Economic Input Output LCA
 - Aggregates the processes into commodities
 - Environmental impacts of each economic activity
- Hybrid Process LCA, EIO LCA
- Dynamic LCA – includes time dependent, temporal impacts

Wrapping it up

- The goal of your LCA is to investigate and compare the environment impact of [the things you are investigating]. The function of [the things you are investigating] is to [state the function]. The LCA will be a [e.g. gate to gate], attributional, consequential and process LCA, which will include [e.g. the construction and use] of the [thing you are investigating]. The environmental impacts investigated in this study are [state the impacts] and the functional unit used is [state the functional unit]. This LCA is for ENGEN370 – Engineering and the Environment.

Looking back at the project brief

- Provide an overview of your product, infrastructure, or process, what it does, how it works, where it is located.
- Develop a specification table for your product, infrastructure, or process.
- Develop a general flow diagram for the manufacture, use and end of life for your product, infrastructure, or process, including inputs and outputs of raw/processed materials, energy, and wastes (quantities not required).
- Review existing literature to see what information/data is available for your product. infrastructure, or process.
- Define the function (i.e. what it does) of your product, infrastructure, or process.
- Define the functional unit by which you can compare your different products, infrastructures, or processes.

Looking back at the project brief

- Select one or two environmental impacts, one of which must be carbon dioxide (or equivalent) emissions. The other impacts can be non-renewable energy consumption, water consumption, or resource depletion, so choose any that align with your project and goals.
- Choose the scope of your LCA as a group, which may include any of the following:
 - Manufacture (e.g. construction of infrastructure or plant, fabrication of your product, construction of a bridge, building, dam)
 - Use (resources used, emissions and wastes from day to day use)
 - End of use (e.g. demolition, recycling, landfill)